

U.G.C. NET Exam

COMPUTER

(Solved with Expl)

1. A basic feasible solution of an $m \times n$ transpiration problem is said to be non-degenerate. if basic feasible solution contains exactly _____ number of individual allocation is _____ positions.
- (a) $m + n + 1$, independent
 - (b) $m + n - 1$, independent
 - (c) $m + n - 1$, appropriate
 - (d) $m - n + 1$, independent

Ans. (b) : Transportation problem is crucial part of linear programming problem which can be connected for required sources of supply to responding destination of demand, with the end goal that the aggregate transpiration cost to be limited

The essential phase of any transportation problem as initial basic feasible solution. Initial basic feasible solution must be feasible i.e. it must satisfy all the supply and demand constraints. The number of positive allocations must be equal to $m + n - 1$ where m is number of rows and n is number of columns.

Non degenerate basic feasible solution : A basic feasible solution is non-degenerate it has exactly $m + n - 1$ positive allocations in individual positions. If the allocations are less than required number, then it is known as degenerate basic feasible solution. This solution is not easy to modify, because it is impossible to draw a closed loop for each occupied cell.

2. Consider the following Linear programming problem (LPP) :

Maximize $z = x_1 + x_2$

Subject to the constraints :

$$x_1 + 2x_2 \leq 2000$$

$$x_1 + x_2 \leq 1500$$

$$x_2 \leq 600$$

$$\text{and } x_1, x_2 \geq 0$$

The solution of the above LPP is :

- (a) $x_1 = 750, x_2 = 750, z = 1500$
- (b) $x_1 = 500, x_2 = 1000, z = 1500$
- (c) $x_1 = 1000, x_2 = 500, z = 1500$
- (d) $x_1 = 900, x_2 = 600, z = 1500$

Ans. (c) : Graphical method can be used. On solving all the constraints, the peaks of the Common area are : A(0, 0) B(0, 600), C(1500, 0), D(1000, 500) z is maximum on all the points of the constraint $x_1 + x_2 \leq 1500$ (as slope of z is same as the constraint)

Now let's try to eliminate the options option 1 and 2 can be eliminated as they are violating 3rd constraint option 4 can be eliminated as it is violating 1st constraint.

. 4 December, 2019

R SCIENCE

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3. The Boolean expression $AB + A\bar{B} + \bar{A}C + AC$ is unaffected by the value of the Boolean variable_____.

- (a) A (b) B
(c) C (d) A,B and C

Ans. (b) : Let $F(A, B, C) = AB + A\bar{B} + \bar{A}C + AC$

$$F(A, B, C) = A(B + \bar{B}) + (\bar{A} + A).C$$

$$F(A, B, C) = A + C \dots (\because A + \bar{A} = 1 \text{ and } 1.A = A)$$

Since the minimized Boolean expression does not contain Boolean variable B. Therefore, $F(A, B, C)$ is unaffected by the value of the Boolean variable B.

4. What are the greatest lower bound (GLB) and the least upper bound (LUB) of the sets $A = \{3, 9, 12\}$ and $B = \{1, 2, 4, 5, 10\}$ if they exist in poset $(z^+, /)$?

- (a) $A(\text{GLB} - 3, \text{LUB} - 36)$; $B(\text{GLB} - 1, \text{LUB} - 20)$
(b) $A(\text{GLB} - 3, \text{LUB} - 12)$; $B(\text{GLB} - 1, \text{LUB} - 10)$
(c) $A(\text{GLB} - 1, \text{LUB} - 36)$; $B(\text{GLB} - 2, \text{LUB} - 20)$
(d) $A(\text{GLB} - 1, \text{LUB} - 12)$; $B(\text{GLB} - 2, \text{LUB} - 10)$

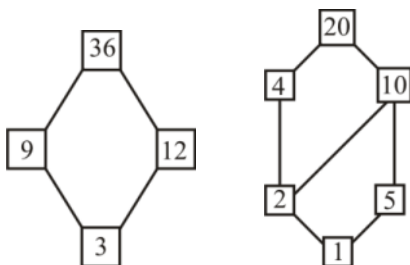
Ans. (a) : In poset $(z^+, /)$, $/$ is the division relation. Hence these diagram for the given POSET

$A = \{3, 9, 12, 36 \text{ (added)}\}$

A with $\text{GLB} = 3$ and $\text{LUB} = 36$

$B = \{1, 2, 4, 5, 10, 20 \text{ (added)}\}$

Both $\text{GLB} = 1$ and $\text{LUB} = 20$



5. Let P be the set of all people. Let R be a binary relation on P such that (a, b) is in R if a is a brother of b . Is R symmetric transitive, an equivalence relation, a partial order relation?

- (a) NO,NO,NO,NO
(b) NO,NO,YES,NO
(c) NO,YES,NO,NO
(d) NO,YES,YES,NO

Ans. (c) : R is a relation on P such that (a, b) is in R if a is a brother of b.

Case 1 : check symmetric relation (a, b) – a is the brother of b,

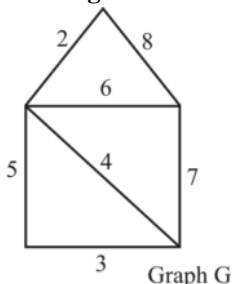
(b, a) – b may or may not be brother of a, so given relation is not symmetric.

Case 2 : Check transitivity If a is the brother of b, b is the brother of c, then a will also be brother of c. It means if (a, b) and (b, c) then (a, c). So given relation is transitive.

Case 3 : Equivalence. As relation is not symmetric, So it cannot be equivalence relation.

Case 4 : Partial order relation for this, we have to check anti symmetric property. i.e. if (a, b) is in relation than (b, a) should not be in relation unless $a = b$, Here it is not partial order relation.

6. The weight of minimum spanning tree in graph G, calculated using Kruskal's algorithm is :

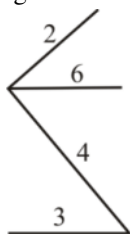


- (a) 14 (b) 15
(c) 17 (d) 18

Ans. (b) : Minimum spanning Tree : → A minimum spanning tree (MST) or minimum weight spanning tree is a subset of the edges of a connected, edge-weighted undirected graph that connects all the vertices together, without any cycles and with the minimum possible total edge weight. So a minimum spanning tree or minimum weight spanning tree is a subset of the edges $(V - 1)$ of a connected, edge weighted undirected graph $G (V, E)$ that connects all the vertices together, without any cycles and with the minimum possible total edge weight.

Edge set for the given graph = {2, 3, 4, 5, 6, 7, 8} for 5 vertices, we need 4 edges in MST. So edge set for MST = {2, 3, 4, 6}

Minimum spanning tree



Minimum cost $\Rightarrow 2 + 3 + 4 + 6 = 15$

7. A tree has $2n$ vertices of degree 1, $3n$ vertices of degree 2, n vertices of degree 3. Determine the number of vertices and edges in tree.
(a) 12,11 (b) 11,12
(c) 10,11 (d) 9,10

Ans. (a) :

Degree of $2n$ vertices = 1

Degree of $3n$ vertices = 2

Degree of n vertices = 3

Formula –

Handshaking theorem : sum of the degree of vertices = $2 \times$ number of edges (E) In a tree, number of edges (E) = number of vertices – 1

Calculation – using 1st formula

$$2n \times 1 + 3n \times 2 + n \times 3 = 2E$$

$$11n = 2E$$

$$E = 2n + 3n + n - 1$$

$$E = 6n - 1$$

Comparing above two equation for E.

$$11n = 12n - 2$$

$$n = 2$$

$$\text{number of vertices} = 6n = 12$$

$$\text{number of edges} = 12 - 1 = 11$$

8. How many reflexive relations are there on a set with 4 elements?

(a) 2^4

(b) 2^{12}

(c) 4^2

(d) 2

Ans. (b) : The number of reflexive relation on an n -Element set is = $2n^2 - n$

Given that set of element (n) = 4

$$= 2n^2 - n$$

$$= 2^{(16-4)} = 2^{12}$$

9. A non-pipelined system takes 30ns to process a task. The same task can be processed in a four-segment pipeline with a clock cycle of 10ns. Determine the speed up of the pipeline for 100 tasks.

(a) 3

(b) 4

(c) 3.91

(d) 2.91

Ans. (d) : For a non-pipelines system.

time to process a task, $t_n = 30\text{ns}$

For a pipelined system.

number of segments, $k = 4$

clock cycle of each segment, $t_p = 10\text{ns}$

For $n = 100$ tasks.

Calculation : $\rightarrow$ Time taken by non pipelined processor to complete 100 tasks = $100 \times$ time taken to complete each task.

$$= 100 \times 30$$

$$= 3000$$

Time taken by pipelined processor to complete 100 tasks = $(100 + \text{number of stages} - 1) \times \text{clock cycle time} = (100 + 4 - 1) \times 10$

$$= 103 \times 10$$

$$= 1030$$

Speed up = (Time taken by non pipelined processor to complete 100 tasks)/ (time taken by pipelined processor to complete 100 tasks)

$$= 3000/1030$$

$$= 2.91$$

10. A computer a memory unit of 512 K works of 32 bits each. A binary instruction code is stored in one word of the memory. The instruction has four parts: an addressing mode field to specify one of the two-address sing mode (direct and indirect), an operation code, a register code part to specify one of the 256 registers and an address part. How many bits are there in addressing mode part, opcode part, register code part and the address part?

- (a) 1, 3, 9, 19
(b) 1, 4, 9, 18
(c) 1, 4, 8, 19
(d) 1, 3, 8, 20

Ans. (c) : Memory unit = 512 k words

1 word = 32 bits = 4B

Binary instruction code stored in 1 word of memory
Instruction divided as follows.

Addressing made	operation code	Register code	Address Part
Addressing mode = 1 bit for direct or indirect, Register code = $\ln(256) = 8$ bits since, registers are already addressed, only memory needs to be addressed. Address part = $\ln(512 \text{ k}) = 19$ bits operation mode = $32 - 1 - 8 - 19 = 4$ bits Therefore, bits for addressing mode part, opcode part, register code part and the address part = (1, 4, 8, 19)			

11. A micro instruction format has micro operation field which is divided into 2 subfields F1 and F2. each having 15 distinct micro operations, condition field CD for four status bits, branch field BR having four options used in conjunction with address field AD. The address space is of 128 memory words. The size of micro instruction is :

- (a) 19 (b) 18
(c) 17 (d) 20

Ans. (a) : MICRO INSTRUCTION FORMAT : →

Micro instruction format consists of two fields of micro - operation. One field from conditions, one for branch field and one is address field.

F1	F2	Condition	Branch	Address
----	----	-----------	--------	---------

Micro operation field into two subfields containing 15 micro operation each.

condition field : 4 status bits

Branch : 4 option in conjunction with address So bits required for micro-operations

$$= \log_2 15 + \log_2 15 = 4 + 4 = 8$$

$$\text{condition} = \log_2 4 = 2$$

Branch in conjunction with address field

$$= \log_2 (4 \times 128) = \log_2 512 = 9$$

F1(4 bits)	F2(4 bits)	condition (2 bits)	Branch with address(9 bits)
------------	------------	--------------------	-----------------------------

size of micro-instruction = 8 + 2 + 9 = 19 bits

12. Given following equation :
 $(142)_b + (112)_{b-2} = (75)_8$, find base b.
 (a) 3 (b) 6
 (c) 7 (d) 5

Ans. (d) : $(142)_b + (112)_{b-2} = (75)_8$
 $b^2 + 4 \times b + 2 + 1 \times (b - 2)^2 + 1 \times (b - 2) + 2 = 7 \times 8 + 5$
 $2b^2 + b + 6 = 61$
 $2b^2 + b - 55 = 0$
 $2b^2 + 11b - 10b - 55 = 0$
 $b(2b + 11) - 5(2b + 11) = 0$
 $(b - 5)(2b + 11) = 0$
 $\therefore b = 5$
Alternate :
Option 1 : $b = 3$
 It has base 3 which is not possible because (142) having a digit 4 cannot exist in base 3
option 2 : $b = 6$ **option 3 :** $b = 7$
 $(142)_6 = (62)_{10}$ $(142)_7 = (79)_{10}$
 $(112)_4 = (22)_{10}$ $(112)_5 = (32)_{10}$
 $62 + 22 = (84)_{10}$ $79 + 32 = (111)_{10}$
 $(75)_8 = (61)_{10}$ $(75)_8 = (61)_{10}$
 There for 6 is not base Therefore 7 is not base
option 4 – $b = 5$
 $(142)_5 = (47)_{10}$
 $(112)_3 = (14)_{10}$
 $47 + 14 = (61)_{10}$
 $(75)_8 = (61)_{10}$
 Therefore 5 is the base
 Hence, $(142)_5 + (112)_3 = (75)_8$

13. The following program is stored in the memory unit of the basic computer. Give the content of accumulator register in hexadecimal after the execution of the program.

Location	Instruction
010	CLA
011	ADD 016
012	BUN 014
013	HLT
014	AND 017
015	BUN 013
016	C1A5
017	93C6

- (a) A1B4 (b) 81B4
 (c) A184 (d) 8184

Ans. (d) :

Location	Instruction	operation	Content of accumulator	order of instruction
010	CLA	Clear Accumulator	0000	1 st
011	ADD 016	ADD (C1A5) to Accumulator	C1A5	2 nd

012	BUN 014	Goto address 014	C1A5	3 rd
013	HLT	Terminate program	C1A5	5 th
014	AND 017	AND (93C6) Accumulator	8184	4 th
015	BUN 013	Goto address 013	8184	
016	C1 A5			
017	93C6			

C1A5 = 1100 0001 1010 0101
& 93C6 = 1001 0011 1100 0110

8184 = 1000 0001 1000 0100

The content of accumulator register in hexadecimal after the execution of the program is – 8184

14. What is the output of following C program?

```
# include <stdio.h>
```

```
main( )
```

```
{
```

```
int i, j, x = 0;
```

```
for (i = 0; i < 5; ++ i)
```

```
for (j = 0; j < i; ++ j)
```

```
{
```

```
x += (i + j - 1);
```

```
break;
```

```
}
```

```
printf ("%d", x);
```

```
}
```

(a) 6

(b) 5

(c) 4

(d) 3

Ans. (a) : Key Point :-

- break statement in the loop the control will come out of that loop only and nothing is executed after the break statement.
- in this question the J- loop will be executed only once as there is break in it.

So totally i will be executed for J = 0 (only) and it will only be executed.

⇒ **Round 1 :**

Initialization :- Value of i = 0, j = 0 and x = 0

As j = 0, j = 0 and j < i

So, no thing will be be printed there.

Round 2 :

i = 1, j = 0 and x = 0

Round 3 :

i = 2, j = 0 and x = 1

Round 4 :

i = 3, j = 0 and x = 3

Round 5 :

i = 4, j = 0, and x = 6

∴ x = 6 is the correct answer.

15. Let A be the base class in C++ and B be the derived class from A with protected inheritance. Which of the following statement is false for class B ?

- (a) Member function of class B access protected data of class A

- (b) Member function of Class can access public data of class A
- (c) Member function of class B cannot access private data of class A
- (d) Object of derived class B can access public base class data

Ans. (d) : Object of Derived class cannot access any member of base class directly. So it cannot access public, protected as well as private member directly. Therefore, object of derived class B can access public base class data is false.

16. Which tag is used to enclose any number of JavaScript statements in HTML document ?

- (a) <code>
- (b) <script>
- (c) <title>
- (d) <body>

Ans. (b) : JavaScript is a programming language for use in HTML pages. Its programs are run by interpreter built into the user's web browser.

programs in JavaScript language are known as scripts. It can execute both on Browser and server or any device that supports special program JavaScript virtual machine. It has full integration with HTML/CSS.

JavaScript Program can be inserted into any part of HTML document with the help of

<script> tag.

Example : <html >

<body>

<script>

alert ("this is JavaScript program");

< / script>

< / body>

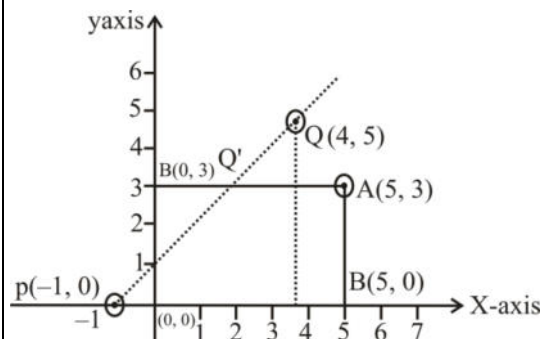
< / html>

17. A rectangle is bound by the lines $x = 0$; $y = 0$; $x = 5$ and $y = 3$. The line segment joining $(-1, 0)$ and $(4, 5)$, if clipped against this window will connect the points _____.

- (a) $(0, 1)$ and $(3, 3)$
- (b) $(0, 1)$ and $(2, 3)$
- (c) $(0, 1)$ and $(4, 5)$
- (d) $(0, 1)$ and $(3, 5)$

Ans. (b) : Given

Rectangle is bound by the lines $x = 0$, $y = 0$, $x = 5$ and $y = 3$



we follow the pattern of

| | | | |
|-----|--------|-------|------|
| Top | Bottom | Right | Left |
|-----|--------|-------|------|

Steps :

1. As p lies in the left of window. So region for this will be : 0001
2. Q point lies in the top of the window, So, region for this will be : 1000
3. Logical AND of 0001 AND 1000 = 0000 i.e. line is partially inside the window.
4. Slope of the line PQ = $(5 - 0) / (4 - (1)) = 5/3 = 1$
5. Now, we have to find the coordinates of 'p', slope of PQ.
6. So, we have to find the Y coordinate as X coordinate is o. $(5 - Y') (4 - 0) = 1, 5 - Y' = 4, y' = 1$ coordinate of p' will be (0, 1).
7. Next we have to find the coordinates of point Q. For this, Y Coordinate we already know i.e. 3.
8. So, $(5 - 3)/(4 - x') = 1, 2 = 4 - x', x' = 2$ coordinates of Point Q' will be (2, 3). So the line segment joining (-1, 0) and (4, 5), if clipped against this window will connect the points (0, 1) and (2, 3).

18. Which of the following algorithms is not used for line clipping?

- (a) Cohen-Sutherland algorithm
- (b) Sutherland-Hodgeman algorithm
- (c) Liang-Barsky algorithm
- (d) Nicholl-Lee-Nicholl algorithm

Ans. (b) : Line clipping : It involves clipping of lines which one outside the window line.

From above given option line clipping algorithms are :

COHEN-SUTHERLAND ALGORITHM : It cuts line to portions which are within a rectangular area. It eliminates those lines which are outside the view port and clip those lines which are partially inside and partially outside of view port. To clip a line, we must have the endpoints. Region code in Cohen- Sutherland algorithm is :

| | | | |
|-----|--------|------|-------|
| TOP | Bottom | Left | right |
|-----|--------|------|-------|

Two cases are there : trivial acceptance and trivial rejection. If both end points are inside the window then trivially accepted. If both end points lie entirely to one side, or outside of window. Then trivially rejected otherwise clip the line.

Liang Barsky Algorithm → It uses four computations to find the appropriate end points of a line. It is based on floating point arithmetic. It uses parametric equations for the line.

Nichole - Lee - Nicholl algorithm : It divides the clipping window into separate areas and matches the start point area code to one of those areas. Imaginary lines are drawn to all the edges of the window and based on these imaginary lines, line is dipped.

19. If we want to resize a 1024 × 768 pixels image to one that is 640 pixels wide with the same aspect ratio, what would be the height of the resized image?

- (a) 420 Pixels (b) 460 Pixels
(c) 480 Pixels (d) 540 Pixels

Ans. (c) : Aspect ratio of an image = width/height Here we have given an image with size 1024×768 pixels.

width = 1024 pixels

Height = 768 pixels

$$\text{Aspect ratio} = \frac{1024}{768} = \frac{4}{3}$$

Also, width changes to 640 pixels which aspect ratio remain same.

$$\frac{4}{3} = \frac{640}{h}$$

$$h = \frac{640 \times 3}{4} = 480$$

So, Height of resized image will be 480 pixels.

20. Which of the component module of DBMS does rearrangement and possible ordering of operations eliminate redundancy in query and use efficient algorithms and indexes during the execution of a query ?

- (a) query compiler
(b) query optimizer
(c) Stored data manager
(d) Database processor

Ans. (b) : There are many plans that a database management system can follow to process a query, but it is the amount of time that they need to run is given importance. For this, query optimization is required.

Query optimization examines all algebraic expressions that are equivalent to the one that is estimated to be the cheapest.

* Query optimization procedure either attempt to maximize of resources or to minimize the resource usage for a given output.

* It tries the minimize the response time for a given query language.

* It does rearrangement and possible ordering of operations, eliminate redundancy in query

* In order to allow fair comparison of efficiency, the functional capabilities of the query evaluation system to be compared must be similar.

21. Given two tables $R1(x, y)$ and $R2(y, z)$ with 50 and 30 number of tuples respectively. Find maximum number of tuples in the output of natural join between tables $R1$ and $R2$ i.e. $R1 \bowtie R2$? (* - Natural Join)

- (a) 30 (b) 20
(c) 50 (d) 1500

Ans. (d) : "A natural Join is based on common attributes or common columns to join two tables or relations

Consider an example, suppose there are two table Employee (Eid, Ename) consists of 5 tuples and department (Eid, Did) consists of 3 tuples only

| Eid | Ename |
|-----|-------|
| 1 | E1 |
| 1 | E2 |
| 1 | E3 |
| 1 | E4 |
| 1 | E5 |

| Eid | Did |
|-----|-----|
| 1 | D1 |
| 1 | D2 |
| 1 | D3 |

Natural Join of employee and department (Employee Department) gives : $5 \times 3 = 15$

So, maximum number of tuples in natural join are from the relation having minimum tuples. So, in given question, R2 contains 30 tuples, So, maximum number of tuples in the output of natural Join between tables R1 and R2 i.e. $R1 * R2 = 50 \times 30 = 1500$.

22. Given two tables EMPLOYEE (EID, ENAME, DEPTNO) DEPARTMENT (DEPTNO, DEPTNAME)

Find the most appropriate statement of the given query :

Select count (*) 'total'

from EMPLOYEE

where DEPTNO IN (D1, D2)

group by DEPTNO

having count (*) > 5

- Total number of employees in each department D1 and D2
- Total number of employees of department D1 and D2 if their total is > 5
- Display total number of employees in both departments D1 and D2
- The output of the query must have atleast two rows

Ans. (b) : Given query is :

```
select count (*) total
from EMPLOYEE where
DEPTNO IN (D1, D2)
group by DEPTNO
having count (*) > 5
```

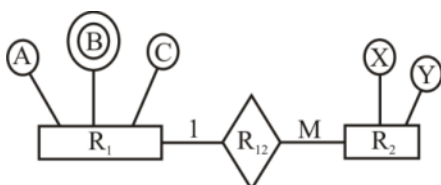
STEP1 :- It will first find the all the entries with department no. as D1 and D2 with the help of natural join from both the relations.

STEP2 :- Then in the resulting table, it groups the tuples with the department no. (DEPTNO)

STEP3 :- Then select all those tuples having count > 5

STEP4 :- Finally, it results in the total number of employees of department D1 and D2 if their total is > 5

23. Find minimum number of tables required for converting the following entity relationship diagram into relational database?



- (a) 2 (b) 4
(c) 3 (d) 5

Ans. (c) : Rules for finding minimum number of tables required for an ER diagram :

1. A strong entity with single or composite attributes require one table.
2. A strong entity with multivalued attribute requires two tables.
3. In case of many to many relations between two entities, 3 tables are required

There is one to many relationships between R1 and R2. So, two tables are required for two entities. But, entity R2 contains multivalued attribute B, due to which one table for this is also needed. So, minimum number of tables required for the given ER diagram are 3.

- 24. A counting semaphore is initialized to 8.3 wait () operations and 4 signal () operations are applied. Find the current value of semaphore variable.**
- (a) 9 (b) 5
(c) 1 (d) 4

Ans. (a) :

V(S) : Signal operation will increment the semaphore variable, that is, $S++$.

P(S) : Wait operation will decrement the semaphore variable that is, $S--$.

Data : Initial counting semaphore = $I = 8$

wait operation = $3P$

signal operation = $4V$

Final (current) counting semaphore = $F \Rightarrow$

Formula : $F = I + 4V + 3P$

Calculation : $F = 8 + 4(+1) + 3(-1)$

$\therefore F = 9$

the resulting value of the semaphore is 2.

- 25. Which of the following is noted by an encryption algorithm used in cryptography?**
- (a) KEY
(b) Message
(c) Ciphertext
(d) User details

Ans. (c and d) : To encrypt any message, any encryption algorithms necessarily require –

1. Plain text (message)
2. Key (private or public)

Encryption algorithm doesn't require cipher text, it produces cipher text. To decrypt any message, any decryption algorithms necessarily require

1. Cipher text
2. key (private or public)

No encryption or decryption algorithm requires user details.

- 26. Which of the following CPU scheduling algorithms is/are supported by LINUX operating system?**

- (a) Non-preemptive priority scheduling
- (b) Preemptive priority scheduling and time sharing CPU scheduling
- (c) Time sharing scheduling only
- (d) Priority scheduling only

Ans. (b) : Linux operation system supports both preemptive priority scheduling and time sharing CPU scheduling. Linux scheduler is a priority preemptive scheduler with two priority ranges-Real time from 0 to 99 and a nice range from 100 to 140.

Preemptive priority scheduling : It maintains two array : active array and expired array. Schedule the processes baseel on their static and dynamic priorities.

1. It finds the highest priority queue with a runnable process and find the first Process on that queue. Calculate its time quantum, it put that process on expire array list and repeat the process again.

2. When priority level of active queue is empty, it looks for next highest queue, after running all active queues, it swaps active and expire queues.

Diagram :

| active
array | | expired
array | |
|-----------------|-----------|------------------|------------|
| Priority | task list | Priority | task lists |
| [0] | 0—0 | [0] | 0—0—0 |
| [1] | 0—0—0 | [1] | 0 |
| • | • | • | • |
| • | • | • | • |
| • | • | • | • |
| [140] | 0 | [140] | 0—0 |

- 27. Consider a paging system where translation look aside buffer (TLB) a special type of associative memory is used with hit ratio of 80%.**

Assume that memory reference takes 80 nanoseconds and reference time to TLB is 20 nanoseconds. What will be the effective memory access time given 80% hit ratio?

- (a) 110 nanoseconds
- (b) 116 nanoseconds
- (c) 200 nanoseconds
- (d) 100 nanoseconds

Ans. (b) : TLB hit ratio = 0.8

TLB miss ration = 0.2

TLB reference time = 20ns

Main Memory reference time = 80ns

Formula : Effective memory access time (EMAT)

= TLB hit (TLB time + Main memory time) + TLB miss (TLB time + 2 × Main memory time)

Calculation :

$$\text{EMAT} = 0.8 (20 + 80) + 0.2 (20 + 80 + 80)$$

$$= 80 + 36$$

$$\therefore \text{EMAT} = 116\text{ns.}$$

28. Suppose a system has 12 magnetic tape drives and at time t_0 , three processes are allotted tape drives out of their need as given below :

Maximum Needs Current Needs

| | | |
|-------|----|---|
| p_0 | 10 | 5 |
| p_1 | 4 | 2 |
| p_2 | 9 | 2 |

At time t_0 , the system is in safe state. Which of the following is safe sequence so that deadlock is avoided ?

- (a) $\langle p_0, p_1, p_2 \rangle$
- (b) $\langle p_1, p_0, p_2 \rangle$
- (c) $\langle p_2, p_1, p_0 \rangle$
- (d) $\langle p_0, p_2, p_1 \rangle$

Ans. (b) : Remaining Need = Maximum need - current need

| | Maximum needs | Current Needs | Remaining Needs |
|----|---------------|---------------|-----------------|
| P0 | 10 | 5 | 5 |
| P1 | 4 | 2 | 2 |
| P2 | 9 | 2 | 7 |

Total Resources = 12
 Total resources allocated = 5 + 2 + 2 = 9
 Available resources = 12 – 9 = 3
Calculation : Available resource $\geq$ remaining need of P1 using 3 resources, P₁ gets executed first. It frees 2 resources.
 Available resources = 3 + 2 = 5
 Available resources $\geq$ remaining need of P₀
 Using 5 resources, P₀ gets executed first. It frees 5 resources.
 Available resources = 5 + 5 = 10
 Available resources $\geq$ remaining need of P2
 Now, P₂ can be executed easily
 Hence, (P₁, P₀, P₂) is a safe sequence.

29. Which of the following interprocess communication model is used to exchange messages among co-operative processes ?

- (a) Shared memory model
- (b) Message passing model
- (c) Shared memory and message passing model
- (d) Queues

Ans. (c) : Inter process communication (IPC) is a mechanism which allows processes to communicate with each other and synchronize their actions. By maintaining this type of communication, processes are said to so - operate with each other. Processes can communicate with each other using these two ways :
 – Shared memory
 – Message passing

30. Given CPU time slice of 2ms and following list of processor.

| Process | Burst time | Arrival time |
|---------|------------|--------------|
| | (ms) | |
| p_1 | 3 | 0 |
| p_2 | 4 | 2 |
| p_3 | 5 | 5 |

Find average turnaround time and average waiting time using round robin CPU scheduling?

- (a) 4,0 (b) 5.66, 1.66
(c) 5.66, 0 (d) 7,2

Ans. (b) : Gantt chart with time slice of 2ms

| | | | | | | |
|----|----|----|----|----|----|----|
| P1 | P2 | P1 | P2 | P3 | P3 | P3 |
|----|----|----|----|----|----|----|

0 2

4 5 7 9 11 12

Process table : (time in milliseconds)

| Pro
cess | Burst
time
(BT) | Arriv
al
time
(AT) | Compl
etion
time
(CT) | Turn
Arou
nd
time
(TAT)
=
(CT-
AT) | waitin
g time
(WT)
=
(TAT-
BT) |
|-------------|-----------------------|-----------------------------|--------------------------------|---|---|
| P1 | 3 | 0 | 5 | 5 | 2 |
| P2 | 4 | 2 | 7 | 5 | 1 |
| P3 | 5 | 5 | 12 | 7 | 2 |
| | | | | Sum
= 17 | Sum =
5 |

Average TAT = $17/3 = 5.66$

Average WT = 1.66

31. Which of the following methods are used to pass any number of parameters to the operating system through system calls?

- (a) Registers
(b) Block or table in main memory
(c) Stack
(d) Block in main memory and stack

Ans. (d) : Following are the methods used to pass parameters to the OS.

1-Registers : Parameters are passed via registers but the limitation of this method is that the number of parameters cannot exceed the number of register.

2-Blocks : when there are more parameters than registers, these parameters are stored on a main memory block and the pointer to this block (block address) is passed as a parameter to a register.

3-Stack : Parameter can also be pushed on and popped off the stack by the OS. This method can be used when we don't certainly know the number of parameters beforehand. Therefore, methods used to pass ANY NUMBER of parameters to the operating system through system calls are via block and stack. Register cannot pass ANY NUMBER of parameters.

32. Java Virtual Machine (JVM) is used to execute architectural neutral byte. Which of the following is needed by the JVM for execution of Java Code?

- (a) Class loader only
(b) Class loader and Java Interpreter
(c) Class loader Java Interpreter and API
(d) Java Interpreter only

Ans. (b): Java virtual machine is component of technology for its hardware and operating system independence, the small size of its compiled code and its ability to protect users from malicious programs.

Java virtual machine is an abstract computing machine, like a real computing machine, it has an instructions set and manipulated various memory areas at run time.

* Java virtual machine knows nothing of java Programming language, only of a particular binary format the class file format.

* A class file contains JVM instructions or byte codes and a symbol table as well as other information.

* JVM imposes strong syntactical and structural constraints on code in a class file

* Class loader and interpreter is needed by JVM to execute Java code

* Basic structure in JVM are : heap store, stack, method area, runtime constant pools, PC

33. In a system for a restaurant, the main scenario for placing order is given below :

- (a) Customer reads menu
- (b) Customer places order
- (c) Order is sent to kitchen for preparation
- (d) Ordered items are served
- (e) Customer requests for a bill for the order
- (f) Bill is prepared for this order
- (g) Customer is given the bill
- (h) Customer pays the bill

A sequence diagram for the scenario will have atleast how many objects among whom the messages will be exchanged .

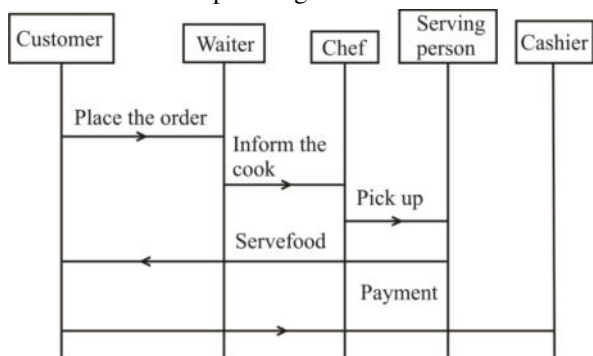
- (a) 3
- (b) 4
- (c) 5
- (d) 6

Ans. (c) : There are five objects among whom the message will be exchanged. :

These are :

1. Customer
2. Person who takes the order
3. Chef
4. Serving person
5. Cashier

Process with the help of diagram is :



34. An _____ chart is a project schedule representation that presents project plan as a directed graph. The critical path is the _____ sequence of _____ tasks and it defines project _____.
- (a) Activity, Shortest, Independent, Cost
 (b) Activity, Longest, Dependent, Duration
 (c) Activity, Longest, Independent, Duration
 (d) Activity, Shortest, Dependent, Duration

Ans. (b) : CPM (critical path method) models the activities and events of a project as a network. Activities are identified as nodes and events that signify the start and end of activities are denoted as edges between nodes.

CPM provides a graphical view of the project. It predicts the time required to complete the project.

* Main aim of CPM is to determine the minimum completion time for a project.

* It uses forward and backward pass routines to analyze the project network.

* It monitors the progress of project on basis of network schedule.

* In CPM, critical path is the longest sequence of dependent tasks and it defines project duration.

* During forward pass, it is assumed that each activity will begin at its earliest starting time.

* During backward pass, it is assumed that each activity begins at its latest completion time.

Steps of CPM project planning :

1. Specification of individual activities and determining the sequence of those activities.
2. Then draw network diagram as directed graph.
3. Estimation of completion time for each activity.
4. Then identify the critical path or longest path through the network.
5. Finally update the CPM as the project progresses.

35. Let $a^{2^c} \bmod n = (a^c)^2 \bmod n$ and $a^{2^{c+1}} \bmod n = a \cdot (a^c)^2 \bmod n$. For $a = 7$, $b = 17$ and $n = 561$. what is the value of $a^b \pmod n$?
- (a) 160
 (b) 166
 (c) 157
 (d) 67

Ans. (a) : $7^{17} \bmod 561$

$$\equiv 7 \left((7^4)^2 \right)^2 \bmod 561$$

$$\equiv 7 \left((2401)^2 \right) \bmod 561$$

$$\equiv 7 \left((157)^2 \right)^2 \bmod 561 \quad [2401 \bmod 561 \equiv 157]$$

$$\equiv (24649)^2 \bmod 561$$

$$\equiv 7(-35)^2 \quad [\because 24649 \bmod 561 \equiv 526 \equiv -35]$$

$$\equiv 7(1225) \bmod 561$$

$$\begin{aligned} &\equiv 7(103) \bmod 561 \\ &\equiv 721 \bmod 561 \\ &\equiv 160 \end{aligned}$$

36. A clique in an undirected graph $G = \langle V, E \rangle$ is a subset $V' \subseteq V$ of vertices, such that

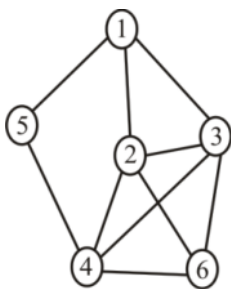
- (a) If $(u, v) \in E$ then $u \in V'$ and $v \in V'$
- (b) If $(u, v) \in E$ then $u \in V'$ or $v \in V'$
- (c) Each pair of vertices in V' is connected by an edge
- (d) All pairs of vertices in V' are not connected by an edge

Ans. (c) : A clique is a subset of vertices of an undirected graph G such that every two distinct vertices in the clique are adjacent

A clique in an undirected graph $G = (V, E)$ is a subset of V i.e. V' such that each pair of vertices in V' is connected by an edge. A clique of size k is called k -clique. A 0 -clique contains set of 0 vertices. A maximal clique is a clique that cannot be extended by including one more adjacent vertex. A maximum clique of G is such that there is no clique with more vertices.

3 cliques :

- 1 – 2 – 3
- 2 – 3 – 4
- 2 – 3 – 6
- 2 – 4 – 6
- 3 – 4 – 6



A Graph with 5 cliques of size 3

37. What is the worst case running time of Insert and Extract-min, in an implementation of a priority queue using an unsorted array? Assume that all insertions can be accommodated.

- (a) $\theta(1)$, $\theta(n)$
- (b) $\theta(n)$, $\theta(1)$
- (c) $\theta(1)$, $\theta(1)$
- (d) $\theta(n)$, $\theta(n)$

Ans. (a) : A priority queue is a queue in which each element is assigned a priority and they are served according to the priority. If element having same priority, they are processed according to their order in queue.

Insert, extract – min is performed on unsorted array.
 Insert : It means insertion of element in unsorted array.
 In the worst case, element is inserted at the end of the

queue. Simply inserting the element at the end of queue takes $O(1)$ complexity.

Extract-min : To do extract-min, use linear search approach for worst case to find the minimum element. For this, we need to traverse all the elements of the queue in worst case. So it will take $O(n)$ time complexity to extract - min in priority queue.

- 38. In a B-Tree, each node represents a disk block. Suppose one block holds 8192 bytes. Each key uses 32 bytes. In a B-tree of order M there are M - 1 keys. Since each branch is on another disk block, we assume a branch is of 4 bytes. The total memory requirement for a non-leaf node is**
- $32 M - 32$
 - $36 M - 32$
 - $36 M - 36$
 - $32 M - 36$

Ans. (b) :

Data :- Block size = 8192 bytes = 2^{13} bytes

Key size = 32 bytes

Block pointer = 4 byte

Order of the b-tree = M

Number of keys = M - 1

FORMULA → In a B - Tree, the size of non - leaf node = M x (Block pointer) + (M - 1) x (Key + Record pointer)

CALCULATION :- Assuming Record pointer = 0 (since nothing is mentioned about it)

Total size = $M(4) + (M - 1)(32) = 36 M - 32$

- 39. Give asymptotic upper and lower bound for $T(n)$ given below. Assume $T(n)$ is constant for $n \leq 2$. $T(n) = 4T(\sqrt{n}) + \lg^2 n$**
- $T(n) = \theta(\lg(\lg^2 n) \lg n)$
 - $T(n) = \theta(\lg^2 n \lg n)$
 - $T(n) = \theta(\lg^2 n \lg \lg n)$
 - $T(n) = \theta(\lg(\lg n) \lg n)$

Ans. (c) : $T(n) = 4T(\sqrt{n}) + \log^2 n$

Let $n = 2^m \Rightarrow m = \log_2 n$

$$\Rightarrow T(2^m) = 4T\left(2^{\frac{m}{2}}\right) + \log^2(2^m)$$

$$\Rightarrow T(2^m) = 4T\left(2^{\frac{m}{2}}\right) + \log(2^m) \times \log(2^m)$$

Now,

$$S(m) = T(2^m)$$

$$\Rightarrow S(m) = 4S\left(\frac{m}{2}\right) + m \times m$$

$$\Rightarrow S(m) = 4S\left(\frac{m}{2}\right) + m^2$$

Now, on applying the

Master's theorem, We get :
 Here, $a = 4, b = 2, k = 2, P = 0$
 $\therefore a = b^k \Rightarrow 4 = 2^2$, and $k = 2, P > 0$
 $\therefore$ **case 2 (i)**
 $T(n) = \Theta (n \log_b a \log^{P+1} n)$
 where $a = b^k, P > -1$
 $\therefore T(n) = \Theta (m^{\log_2 4} \log^{0+1} m)$
 $= \Theta (m^{\log_2 2^2} \log^{0+1} m)$
 $= \Theta (m^{2 \log_2 2} \log^{0+1} m)$
 $= T(n) = \Theta (m^2 \times \log^1 m) \dots (1)$
 Now $\therefore m = \log_2 n$
 Substituting this value of m in (1), we get :
 $T(n) = \Theta (\log^2 n. \log n)$
 So, option 3 is the right answer.

- 40. Consider a weighted directed graph. The current shortest distance from source S to node x is represented by $d[x]$. Let $d[v] = 29, d[u] = 15, w[u, v] = 12$. What is the updated value of $d[v]$ based on current information?**
- (a) 29 (b) 27
 (c) 25 (d) 17

Ans. (b) : If d represents the length of the path, then :
 $d(v) = \begin{cases} 0, & \text{if } v = s \\ \min_{u: (u,v) \in E} \{d(u) + w(u, v)\}, & \text{otherwise} \end{cases}$
 where $w(u, v)$ is the weight of the edge (u, v)
 $\therefore$ Here
 $d(u) + w(u, v) = 15 + 12 = 27$
 $\therefore 27$ is the correct answer.

- 41. When using Dijkstra's algorithm to find shortest path in a graph, which of the following statement is not true?**
- (a) It can find shortest path within the same graph data structure
 (b) Every time a new node is visited, we choose the node with smallest known distance/cost (weight) to visit first
 (c) Shortest path always passes through least number of vertices
 (d) The graph needs to have a non-negative weight on every edge

Ans. (c) :

Dijkstra's algorithm is used to find single source shortest path i.e. path between any two vertices in a graph. All edges must have non-negative weights and graph must be connected.

| v | a | b | c | d | e | f | g |
|---|-------|-------|-------|-------|--------|-------|--------|
| a | 0_a | 3_a | 5_a | 6_a | Inf | Inf | Inf |
| b | 0_a | 3_a | 5_a | 5_b | Inf | Inf | Inf |
| c | 0_a | 3_a | 5_a | 5_b | 11_c | 8_c | 12_c |
| d | 0_a | 3_a | 5_a | 5_b | 11_c | 8_c | 12_c |
| | 0_a | 3_a | 5_a | 5_b | 11_c | 8_c | 9_f |

So, we get the shortest distance from a to g. ($a \rightarrow b \rightarrow c \rightarrow d \rightarrow f \rightarrow g$).

42. The time complexity to multiply two polynomials of degree n using Fast Fourier transform method is:

- (a) $\theta(n \lg n)$ (b) $\theta(n^2)$
 (c) $\theta(n)$ (d) $\theta(\lg n)$

Ans. (a) : Fast Fourier transform is a discrete Fourier transform algorithm which reduces the number of computation needed for N points from $2N^2$ to $2N \log N$.

A discrete Fourier transform can be computed using fast Fourier transform by means of Daniel-Sonlanczos lemma, if the number of point N is a Power of two.

If number of points N is not a power of two, a transform can be performed n sets of points corresponding to prime factor of N.

Fast Fourier transform allows computing in $\theta(n \log n)$ time. Basic idea is to apply divide and conquer. We divide the co-efficient vector of polynomial into two vectors, recursively compute the Discrete Fourier transform for each of them and combine the results Recursive relation will become;

$$T_f(N) = 2T_f(N/2) + O(N).$$

43. Consider the following grammars:

- $G_1: S \rightarrow aSb|bSa|aa$
 $G_2: S \rightarrow aSb|bSa|SS|\lambda$
 $G_3: S \rightarrow aSb|bSa|SS|a$
 $G_4: S \rightarrow aSb|bSa|SS|SSS|\lambda$

Which of the following is correct w.r.t the above grammars?

- (a) G_1 and G_3 are equivalent
 (b) G_2 and G_3 are equivalent
 (c) G_2 and G_4 are equivalent
 (d) G_3 and G_4 are equivalent

Ans. (c) : $G_1: S \rightarrow aSb|bSa|aa$

Gives a language $G_1 = \{ \text{number of a} = \text{number of b} + 2 \}$

$$G_2: S \rightarrow aSb|bSa|SS|\lambda$$

Gives a language $G_2 = \{ \text{number of a} = \text{number of b} \}$

$$G_3: S \rightarrow aSb|bSa|SS|a$$

Gives a languages $G_3 = \{ \text{number of a} \geq \text{number of b} + 1 \}$

$G_4 : S \rightarrow aSb|bSa|SS|SSS|\lambda$
 Gives a language $G_4 = \{\text{number of } a = \text{number of } b\}$
 Hence, G_2 and G_4 are equivalent.

- 44. Consider the language $L = \{a^n b^{n-3} \mid n > 2\}$ on $\Sigma = \{a, b\}$. Which one of the following grammars generates the language L ?**
- (a) $S \rightarrow aA|a, A \rightarrow aAb|b$
 - (b) $S \rightarrow aaA|\lambda, A \rightarrow aAb|\lambda$
 - (c) $S \rightarrow aaaA|a, A \rightarrow aAb|\lambda$
 - (d) $S \rightarrow aaaA, A \rightarrow aAb|\lambda$

Ans. (d) : for $n = 3$, string generated = aaa
 for $n = 4$, string generated = aaaab
 for $n = 5$ string generated = aaaaabb
 only option 4 can generate the above grammar at following value of n .

- 45. Consider the following grammar :**
- $S \rightarrow 0A|0BB$
 $A \rightarrow 00A|\lambda$
 $B \rightarrow 1B|11C$
 $C \rightarrow B$
- Which language does this grammar generate ?**
- (a) $L((00)^*0 + (11)^*1)$
 - (b) $L(0(11)^*0 + 1(00)^*)$
 - (c) $L((00)^*0)$
 - (d) $L(0(11)^*1)$

Ans. (c) : Given grammar is generating string of type:
 $\{0, 000, 00000, 0000000, \dots\}$

(1) $L((00)^*0 + (11)^*1)$
 This language generating string with combination of 0's and 1 and But given Grammar is not generating any string composed of 1's.

(2) $L(0(11)^*1(00)^*)$
 this language is not generating string which contains three o's together. Also, it is generating string which contains 1 in it. so, it is not correct option .

(3) $L((00)^*0)$
 It is generating string of type $L:\{0, 0^3, 0^5, 0^7\}$ these set of string is similar to the one generating from given grammar so, this language is generating given grammar.

(4) $L(0(11)^*1)$
 It is incorrect because it is generating string which contains 1 which is not the case with given grammar.

- 46. Consider $\Sigma = \{w, x\}$ and $T = \{x, y, z\}$. Define homomorphism h by:**
- $h(x) = xzy$
 $h(w) = zxyy$
- If L is the regular language denoted by $r = (w+ x^*)(ww)^*$, then the regular language $h(L)$ is given by**
- (a) $(zxyy + xzy)(zxyy)$
 - (b) $(zxyy+(xzy)^*)(zxyyzxyy)^*$

(c) $(zxyy+xzy)(zxyy)^*$

(d) $(zxyy+(xzy)^*)(zxyyzxyy)$

Ans. (b) : Homomorphism is a function from string to string that is based on the concatenation for any x and $y \in \Sigma^*$ $h(xy) = h(x)h(y)$

L is the regular language with the regular expression $r = (w + x^*)(ww)^*$

Then $h(L) = (h(w) + h(x)^*)(h(w)h(w))^*$

Given : $h(x) = xzy$

$h(w) = zxyy$

$h(L) = (zxyy+(xzy)^*)(zxyyzxyy)^*$

47. Consider the following statements with respect to the language $L = \{a^n b^n \mid n \geq 0\}$

$S_1 : L^2$ is context free language

$S_2 : L^k$ is context free language for any given $k \geq 1$

$S_3 : \bar{L}$ and L^* are context free language

Which one of the following is correct?

(a) Only S_1 and S_2

(b) Only S_1 and S_3

(c) Only S_2 and S_3

(d) S_1, S_2 and S_3

Ans. (d) : $L = \{a^n b^n \mid n \geq 0\}$

The given language is context free language and CFL are closed under Union concatenation, and kleen closure.

$S_1 : L^2$ is a context free language, here L^2 can be written as $L^2 = L.L$ and CSL is closed under concatenation so L^2 is a CFL

$S_2 : L^k$ context free language for any given $k \geq 1$, L^k can be written as $L.L.L \dots$ k times and CFL are closed under concatenation so L^k is CFL

$S_3 : \bar{L}$ and L^* are context free languages are not closed under complementation for complement of L may be CFL or CSL

Here complement of L is CFL, and CFL are closed under kleen Closure (*) so it is and CFL.

48. Consider the following languages:

$L_1 = \{a^n b^n c^m\} \cup \{a^n b^m c^m\}, n, m \geq 0$

$L_2 = \{ww^R \mid w \in \{a, b\}^*\}$, where R represents reversible operation.

Which one of the following is (are) inherently language(s) ?

(a) Only L_1

(b) Only L_2

(c) Both L_1 and L_2

(d) Neither L_1 nor L_2

Ans. (a) : L_1 is inherently ambiguous language because we cannot have any unambiguous grammar for this language.

Specifically (Hopcroft & Ullman(1979) gives a proof) that there is no way to unambiguously parse string in the (non-context-free) common subset $\{a^n b^n c^n\}$.

L_2 is not inherently ambiguous and one unambiguous grammar for this language is: $S \rightarrow aSa|bSb| \epsilon$

It is not hard to show that the above grammar is unambiguous (Hint: At every symbol, while reading the string from the left, you only have one choice of production rule in the parse tree.)

49. Let $G = (V, T, S, P)$ be any context - free grammar without any λ -productions or unit production. Let K be the maximum number of symbols on the right of any production in P . The maximum number of production rules for any equivalent grammar in Chomsky normal form is given by:

- (a) $(K-1)|P| - |T| - 1$
- (b) $(K-1)|P| + |T|$
- (c) $K|P| + |T| - 1$
- (d) $K|P| + |T|$

Ans. (b) : Context free Grammar :- A grammar $G = (V, T, S, P)$ is said to be context- free if all productions in P have the form $A \rightarrow X$ Where $A \in V$ and $X \in (V \cup T)^*$

If context the grammar $G = (V, T, S, P)$ is without λ -productions or unit productions.

K = maximum number of symbols on right-side of production.

The maximum number of production rules for equivalent grammar in CNF: $(K-1)|P| + |T|$

A context free grammar is in CNF (Chomsky normal form) if it satisfies these conditions:

1. Should not contains null or unit productions.
2. A non- terminal symbol generating two non-terminals for examples $S \rightarrow AB$
3. A non-terminal generating a terminal example: $S \rightarrow a$

50. Consider the following language families:

$L_1 \equiv$ The context – free languages

$L_2 \equiv$ The context – sensitive languages

$L_3 \equiv$ The recursively enumerable languages

$L_4 \equiv$ The Recursive languages

Which one of the following option is correct?

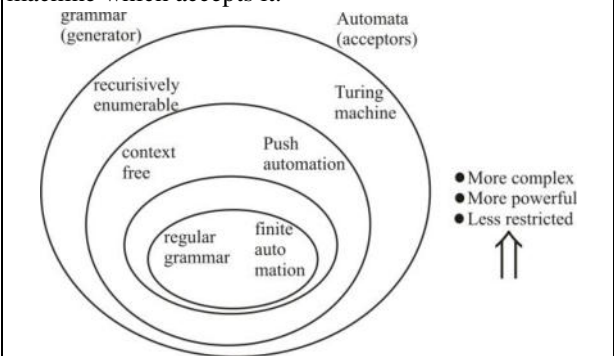
- (a) $L_1 \subseteq L_2 \subseteq L_3 \subseteq L_4$
- (b) $L_2 \subseteq L_1 \subseteq L_3 \subseteq L_4$
- (c) $L_1 \subseteq L_2 \subseteq L_4 \subseteq L_3$
- (d) $L_2 \subseteq L_1 \subseteq L_4 \subseteq L_3$

Ans. (c) : Context free languages:- A language is said to be context free if and only if there exists a context free grammar for it. A grammar $G = (V, T, S, P)$ is said to be context free if all production is in the form of $A \rightarrow X$ where $A \in V$ and $X \in (V \cup T)^*$. There are accepted by push down

Context sensitive languages:- If there exists a context sensitive grammar for a language then it is known context sensitive language context sensitive grammar is one in which all production are of the form $x \rightarrow y$ where $x, y \in (V \cup T)^*$ and $|x| \leq |y|$ these are accepted by linear bounded automata

Recursive language:- A language is recursive if there exists a Turing machine accepts it and halts on every input string .

Recursive enumerable languages : A language is recursive enumerable if there exists some Turing machine which accepts it.



51. Consider the following statements:

S_1 : These exists no algorithm for deciding if any two Turing machines M_1 and M_2 accept the same language.

S_2 : Let M_1 and M_2 be arbitrary Turing machines. The problem to determine $L(M_1) \subseteq L(M_2)$ is undecidable.

Which of the statement is (are) correct?

- (a) Only S_1
- (b) Only S_2
- (c) Both S_1 and S_2
- (d) Neither S_1 nor S_2

Ans. (c) : S_1 There exists no algorithms for deciding if any Turing machine M_1 and M_2 accept the same language"

Statement S_1 is correct because "Equivalence problem of recursively enumerable languages is undecidable.

Hence, No algorithm possible

S_2 : "The problem of determining whether a Turing machine halts on any input is undecidable".

Statement S_2 is correct because subset problem of recursively enumerable languages is undecidable. It is standard "Halting problem of TM". Hence, no algorithm possible.

52. Let $A = \{001,0011,11,101\}$ and

$B = \{01,111,111,010\}$. Similarly, Let

$C = \{00,001,1000\}$ and $D = \{0,11,011\}$.

Which of the following pairs have a post-correspondence solution?

- (a) Only pair (A,B)
- (b) Only Pair(C,D)
- (c) Both (A,B) and (C,D)
- (d) Neither (A,B) nor (C,D)

Ans. (a) : POST CORRESPONDENCE PROBLEM (PCP): It defines the un-decidability of strings for the solution of PCP, we have to check whether there is a

finite sequence $(i_1, \dots, i_p)$ with i_j belongs to $\{1, \dots, m\}$ for $J = 1, \dots, P$ so that

$$U_{i1}U_{i2}\dots U_{ip} = V_{i1} V_{i2}\dots V_{ip}.$$

CASE : for pair (A, B)

A: {001, 0011, 11, 101}

B: {01, 111, 111, 010}.

Consider it as: $A_1 = 001, A_2 = 0011, A_3 = 11, A_4 = 101$

$B_1 = 01, B_2 = 111, B_3 = 111, B_4 = 010$

Here $A_3 A_4 = B_2 B_1$ or $A_3 A_4 = B_3 B_1$ i.e $11101 = 11101$

So pair (A, B) has a PCP solution.

Case 2: Consider pair (C, D)

C = {00, 001, 1000}

D = {0, 11, 011}

There does not exist any such combination of strings which satisfy the property for PCP solution. So pair (C, D) does not have PCP solution.

53. Which of the following class of IP address has the last address as 223.255.255.255?

- (a) Class A (b) Class B
(c) Class C (d) Class D

Ans. (c) : Class A 0.0.0.0 to 127.255.255.255

Class B 128.0.0.0 to 191.255.255.255

Class C 192.0.0.0 to 223.255.255.255

Class D 224.0.0.0 to 239.255.255.255

Class E 240.0.0.0 to 255.255.255.255

So Class of IP address 223.255.255.255 is belongs to class C.

54. Consider a subnet with 720 routers. If a three-level hierarchy is chosen. with eight clusters, each containing 9 regions of 10 routers. Then total number of entries in hierarchical table of each router is.

- (a) 25 (b) 27
(c) 53 (d) 72

Ans. (a) : In link state and distance vector routing. Every router has to save some information about other routers. When network size grows, the number of routers in network increases. Due to this, routers can't handle network traffic efficiently. That's why there comes a need of hierarchical, routing.

In hierarchical routing, routers are classified in groups Known as regions in three level hierarchical routing, network is classified into a number of clusters. Each cluster is made up of number of regions and each region contain number of routers.

Given: number of routers = 720

clusters = 8

Regions = 9 to 10 routers

Total number of entries in hierarchical table of each router = (number of cluster – 1) + (number of region – 1) + routers in each region.

$$= (8 - 1) + (9 - 1) + (10) = 7 + 8 + 10 = 25$$

55. Piconet is a basic unit of a bluetooth system consisting of — master node and up to — active slave nodes.

- (a) One, five (b) One, seven
(c) two, eight (d) one, eight

Ans. (b) : Bluetooth is a wireless LAN technology designed to connect devices of different functions. A blue tooth LAN is an ad hoc network which means that the network is formed spontaneously, devices sometimes called gadgets, find each other and make a network called piconet.

Bluetooth defines two types of network: piconet scatternet

PICONET→ A Bluetooth network is called, piconet or small net. A piconet can have up to eight stations one of which is called primary and rest one secondary's. All the secondary stations synchronize their clocks and hopping sequence with the primary. Communication between primary and secondary can be one to one or one to many. A piconet can have a maximum of 7 secondaries.

56. A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

- (a) 1,000,000 bps (b) 2,000,000 bps
(c) 12,000,000 bps (d) 1,200,00,000 bps

Ans. (b) : Throughput is amount of data moved successfully from one place to another place i.e 12,000 frames per minute.

Each frame is carrying 10,000 bits

So $12,000 * 10,000 / 60$ seconds.

$= 2 * 1000000$ bits/seconds

$= 2000000$ bps.

57. The full form of ICANN is

- (a) Internet Corporation for Assigned Name and Numbers
(b) Internet Corporation for Assigned Number and Name
(c) Institute of Corporation for Assigned Name and Numbers
(d) Internet Connection for Assigned Name and Numbers

Ans. (a) : ICANN → Internet corporation for assigned Names and Numbers.

58. According to Dempster-Shafer theory for uncertainty management.

- (a) $Bel(A) + Bel(\neg A) \leq 1$
(b) $Bel(A) + Bel(\neg A) \geq 1$
(c) $Bel(A) + Bel(\neg A) = 1$
(d) $Bel(A) + Bel(\neg A) = 0$

Where $Bel(A)$ denotes Belief of event A.

Ans. (a) : Dempster:- Shafer theory: It is designed to deal with the distinction between uncertainty and ignorance Rather than computing the probability of a proposition it computes the probability that evidence

supports the proposition. Rather than estimating probabilities it uses belief intervals to estimate how close the evidence is to determining the truth of a hypothesis. It allows representation of ignorance about support provided by evidence.

* Belief or possibility is the probability that B is provable by the evidence.

* Plausibility is the probability that B is compatible with the available evidence.

* According to Dempster- Shafer theory $(\text{Bel } A) = 1 - \text{Bel } (\neg A)$

for example: if there are three groups A, B and C who did the crime. We may have evidence the group.

C is guilty. We would not want to say the probability of other two groups being guilty is 1. Belief and Disbelief are functional opposites.

Dempster- Shafer theory allows to leave relative beliefs unspecified.

59. Consider the following statements:

$S_1 : \forall x P(x) \vee \forall x Q(x)$ and $\forall x P(x) \vee Q(x)$ are not logically equivalent.

$S_2 : \exists x P(x) \wedge \exists x Q(x)$ and $\exists x P(x) \wedge Q(x)$ are not logically equivalent.

Which of the following statements is/are correct?

- (a) Only S_1
- (b) Only S_2
- (c) Both S_1 and S_2
- (d) Neither S_1 and S_2

Ans. (c) : Here we are considering the example to prove that given statements are true:

Let $p(x)$ be the statement that x is an even number and

Let $Q(x)$ be the statement that x is an odd number, where x denotes the domain of all positive integers.

$\therefore D = N = \text{natural numbers and}$

$p(x) = "x \text{ is even}"$

$Q(x) = "x \text{ is odd}"$

First prove that statement, S_1 is true

$\forall x P(x) \vee \forall x Q(x)$ and $\forall x (p(x) \vee Q(x))$ are not logically equivalent.

Because here, all positive numbers are not even.

So, $(\forall x p(x))$ is false and all positive number are not odd. So, $(\forall x Q(x))$ is false

So, $\forall x p(x) \vee \forall x Q(x)$ is false But $\forall x (p(x) \vee Q(x))$ will always be true, because x will always be either even or odd.

Hence, $\forall x p(x) \vee \forall x Q(x)$ and $\forall x (p(x) \vee Q(x))$ are not logically equivalent.

Now prove that statement, S_2 is true $\rightarrow \exists x p(x) \wedge \exists x$

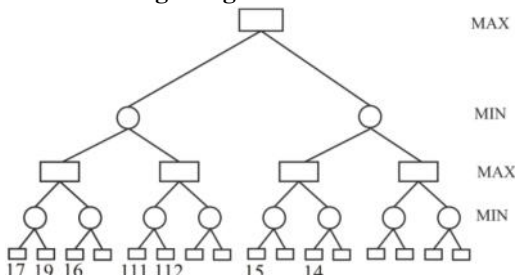
$Q(x)$ and $\exists x (p(x) \wedge Q(x))$ are not logically equivalent.

Because here, there exist some positive number that is even.

So, $\exists x p(x)$ is true and there exist some positive number that is odd. So, $\exists x p(x)$ is true.

So, $\exists x p(x) \wedge \exists x Q(x)$ is true. But $\exists x(p(x) \wedge Q(x))$ and $\exists x(p(x) \wedge Q(x))$ are not logically equivalent.

60. Consider the game given below

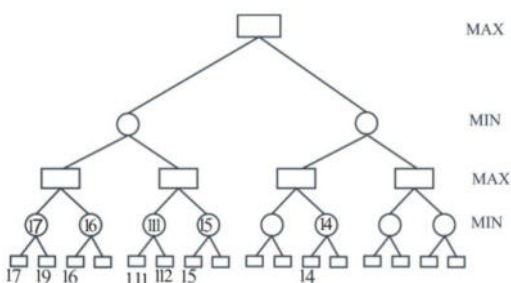


Here $\bigcirc$ and $\square$ represent MIN and MAX nodes respectively. The value of the root node of the game tree is:

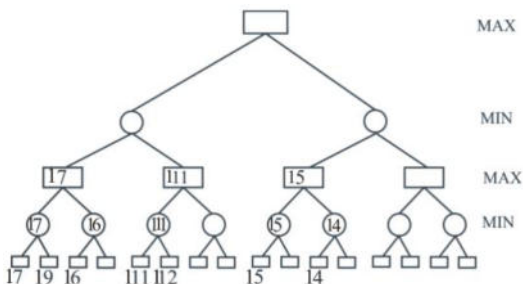
- (a) 14 (b) 17
(c) 111 (d) 112

Ans. (b) : start from bottom to top:

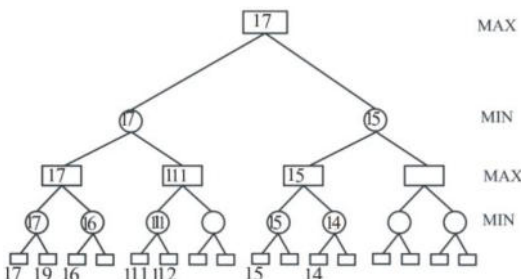
1. first it finds the minimum value from the given values:



2. At next step, max value is considered.



and in then last



61. Consider the following models;
M₁ :Mamdani model
M₂ :Takagi-Sugeno-Kang model
M₃ :Kosko's additive model (SAM)
Which of the following option contains examples of additive rule model ?
 (a) Only M₁ and M₂ (b) Only M₂ and M₃
 (c) Only M₁ and M₃ (d) M₁,M₂ and M₃

Ans. (b) : Mamdoni Model: In a Mamdani system, output of each rule is fuzzy set. These are well suited to expert system applications where the rules are created from human expert knowledge. It is a method to create a control system by synthesizing a set of linguistic control rules obtained from experienced human operators.

Takagi-Sugeno-kang Model: This model allows to construct the system of analysis from a set of example data. These are used for the approximation of a large class of non linear systems. The degree of relevance of a linear model is determined by the degree the input data belong to fuzzz subspace associated with the linear model. This is an example of additive rule model.

Kosko's additive model (SAM): It assumes the inputs are crisp and use the scaling method. It uses addition to combine the conclusion of fuzzy rules and includes the centroid defuzzification technique.

62. A fuzzy conjunction operators , **t(x, z)**and a fuzzy disjunction operator , **s(x, y)**, from a pair if they satisfy:
- t(x, y) = 1 –s(1–x, 1–y)**
- If **t(x, y) = $\frac{xy}{(x + y - xy)}$** then **s(x, y)** is given by
- (a) $\frac{x + y}{1 - xy}$ (b) $\frac{x + y - 2xy}{1 - xy}$
 (c) $\frac{x + y - xy}{1 - xy}$ (d) $\frac{x + y - xy}{1 + xy}$

Ans. (b) : s(x, y)→ $\frac{x + y - 2xy}{1 - xy}$

63. Let **W_{ij}** represents weight between node **i** at layer **k** and node **j** at layer **(k–1)** of a given multilayer perceptron. The weight updating using gradient method is given by
- (a) **W_{ij} (t + 1) = W_{ij}(t) + α $\frac{\partial E}{\partial W_{ij}}$, 0 ≤ α ≤ 1**
 (b) **W_{ij} (t + 1) = W_{ij}(t) – α $\frac{\partial E}{\partial W_{ij}}$, 0 ≤ α ≤ 1**
 (c) **W_{ij} (t – 1) = α $\frac{\partial E}{\partial W_{ij}}$, 0 ≤ α ≤ 1**
 (d) **W_{ij} (t + 1) = – α $\frac{\partial E}{\partial W_{ij}}$, 0 ≤ α ≤ 1**

Where **α** **E** represents learning rate and Error in the output respectively.

$$\text{Ans. (b) : } W_{ij}(t+1) = W_{ij}(t) - \alpha \frac{\partial E}{\partial W_{ij}}, 0 \leq \alpha \leq 1$$

64. The order of Schema ?10?101? and ???0??1 are —— and —— respectively .

- (a) 5,3 (b) 5,2
(c) 7,5 (d) 8,7

Ans. (b) : A schema is a set of binary strings that match the template which is made up of I's, 0's and * s where * matches with either 0 or 1. The order of schema is the number of its fixed bits i.e number of bits that are either 0 or 1.

Explanation:

Given schema is: ?10? 101? and ??? 0??1

CASE 1:

FOR ? 10? 101?, number of bits that are either 0 or 1 are 5.

So order of this schema is 5.

CASE 2: FOR ???0??1, number of bits that are fixed are 2.

So order of schema is 2.

65. Let the population of chromosomes in genetic algorithm is represented in terms of binary number. The strength of fitness of a chromosome in decimal form, x, is given by

$$S f(x) = \frac{f(x)}{\sum f(x)} \text{ where } f(x) = x^2$$

The population given by P where:

P = {(01101), (11000), (01000), (10011)}

The strength of fitness of chromosome (11000) is _____.

- (a) 24 (b) 576
(c) 14.4 (d) 49.2

Ans. (d) : In this question given that The population of chromosome in genetic algorithm is represent in term of binary number. The strength of fitness of chromosome in decimal form of x

The population is given by $p = \{01101, (11000), (01000), (10011)\}$

| P | X-value | $f(x) = X \wedge 2$ |
|-------|---------|---------------------|
| 01101 | 13 | 169 |
| 11000 | 24 | 576 |
| 01000 | 8 | 64 |
| 10011 | 19 | 361 |
| SUM | | 1170 |

$$s f(x) = f(x) / \sum f(x) \text{ where } f(x) = X \wedge 2$$

strength of fitness of chromosome (11000) is = $f(11000) / f(01101) + f(11000) + f(01000) + f(10011) =$

$$= 576 / 169 + 576 + 64 + 361$$

$$= 576 / 1170 = 0.492$$

$$= 0.492 \times 100 = 49.2$$

S₁: If a group $(G, *)$ is of order n , and $a \in G$ is such that $a^m = e$ for some integer $m \leq n$ Then m must divide n .

S₂: If a group $(G, *)$ is of even order, then there must be an element $a \in G$ such that $a \neq e$ and $a * a = e$

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) Neither S_1 nor S_2

Given statement is correct. As $a \in G$, is such that $a^m = e$ for some integer $m \leq n$, then it means it is a subgroup of G and m is the order of a , where order of given group is n . By the property of subgroup or Lagrange's theorem, a order of subgroup divides the order of a group. So, here m must divide n is true.

This statement is correct. Consider an example for this: consider G is of order $2n$. There exist $a \in G$ such that $a^p = e$ and p divides $2n$. Let $n = p \cdot q$. So, $(a^n)^2 = (a^{pq})^2 = ((a^p)^q)^2 = (e^q)^2 = e$.

67. Consider the following statements with respect to duality in LPP:

- (A) The final simplex table giving optimal solution of the primal also contains optimal solution of its dual in itself.
- (B) If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution.
- (C) If either problem has an unbounded optimum solution, then the other problem has no feasible solution at all.

(a) only (A) and (B)
(b) only (A) and (C)
(c) only (B) and (C)
(d) (A), (B) and (C)

Some points about dual LPP.

- (1) In final simplex table giving optimal solution of the primal also contains optimal solution of its dual in itself.
- (2) If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution.

(3) If either problem has an unbounded optimum solution, then the other problem has no feasible solution at all.

Example- We have a set of equations as:

$$\text{Maximize } z = 10X_1 + 20X_2$$

Subject to:

$$3X_1 + 4X_2 \leq 100$$

$$5X_1 + 6X_2 \leq 150$$

$$X_1, X_2 \geq 0$$

Dual of this will be:

$$\text{Minimize } z = 100Y_1 + 150Y_2$$

$$\text{Subject to : } 3Y_1 + 4Y_2 \geq 10$$

$$5Y_1 + 6Y_2 \geq 20$$

$$Y_1, Y_2 \leq 0$$

68. The Reduced Instruction Set Computer (RISC) characteristics are:

(A) Single cycle instruction execution

(B) Variable length instruction formats

(C) Instruments that manipulates operands in memory

(D) Efficient instruction pipeline

Choose the correct characteristics from the options given below:

(a) (A) and (B)

(b) (B) and (C)

(c) (A) and (D)

(d) (C) and (D)

Ans. (c) : Statement a : (Correct) RISC has pipelined implementations with the goal of executing one instruction per machine cycle.

Statement b : (Incorrect) RISC has fixed length instruction formats.

Statement c : (Incorrect) C-RISC can perform only register to register arithmetic operations.

Statement d : (Correct) D-RISC has efficient instruction pipeline therefore, statement a and b are correct.

69. Which of the following binary codes for decimal digits are self complementing?

(A) 8421 code

(B) 2421 code

(C) excess-3 code

(D) excess-3 gray code

Choose the correct option:

(a) (A) and (B)

(b) (B) and (C)

(c) (C) and (D)

(d) (D) and (A)

Ans. (b) : A binary coded decimal code in which of the combination for the complement of a digit is the complement of the combination for that digit.

Self-complementing codes are : excess-3 code 2421 cod.

(1) 2421 code- It is positive weighted code values in binary terms for each digit from 0 to 9 are given as:

| Decimal Digit | 2421code |
|---------------|----------|
| 0 | 0000 |
| 1 | 0001 |
| 2 | 0010 |
| 3 | 0011 |
| 4 | 0100 |
| 5 | 1011 |
| 6 | 1100 |
| 7 | 1101 |
| 8 | 1110 |
| 9 | 1111 |

(2) Excess-3 code: It is a non-weighted code and self complementary code. It is derived from 8421 code by adding 3 to each digit.

| Decimal Digit | Excess – 3code |
|---------------|----------------|
| 0 | 0011 |
| 1 | 0100 |
| 2 | 0101 |
| 3 | 0110 |
| 4 | 0111 |
| 5 | 1000 |
| 6 | 1001 |
| 7 | 1010 |
| 8 | 1011 |
| 9 | 1100 |

70. Consider the following statements with respect to approaches to fill area on raster systems:

- (A) To determine the overlap intervals for scan lines that cross the area.**
- (B) To start from a given interior position and paint outward from this point until we encounter the specified boundary conditions.**

Select the correct answer from the options given below:

- (a) (A) only
- (b) (B) only
- (c) Both (A) and (B)
- (d) Neither (A) nor (B)

Ans. (c) : In raster scan system, electron beam move across the screen one row at a time from top to bottom. In this electron beam intensity is turned on and off to produce illumination on the screen.

There are two approaches used in area filling on raster systems: one is scan line algorithm and another is seed fill algorithm.

(1) Scan line algorithm: Determine the overlap intervals for scan lines that cross the area or scan line. A scan line is a line of pixels that are plotted on display device when electron beam scans the display area horizontally.

(2) Seed fill algorithm: It starts from interior position and point outward from this point until the boundary condition reached. In this a point is selected and then four or eight connected approaches is used to fill it.

71. Which of the following statements are true regarding C++?

(A) Overloading gives the capability to an existing operator to operate on other data types.

(B) Inheritance in object oriented programming provides support to reusability.

(C) When object of a derived class is defined first the constructor of derived class is executed then constructor of a base class is executed.

(D) Overloading is a type of polymorphism.

Choose the correct option from those given below:

(a) (A) and (B) only

(b) (A), (B) and (C) only

(c) (A), (B) and (D) only

(d) (B), (C) and (D) only

Ans. (c) : Statement (a) : Overloading gives the capability to an existing operator to operate on other data types.

This statement is incorrect. Operator overloading is a type of polymorphism in which an operator is overloaded to give a different meaning or user defined meaning. It allows to change the way an operator can work. It gives the capability to an existing operator to operate on other data types.

Statement (b): Inheritance in object oriented programming provides support to reusability.

This statement is correct. When a class acquires the properties of another class, this is known as inheritance. The class which acquire the properties and method is known as derived class and other one is parent or base class. Inheritance another name for reusability.

Statement (c): When object of a derived class is defined, first the constructor of derived class is executed then constructor of a base class is executed then constructor of a base class is executed.

This statement is incorrect constructor is a special member function of class that is automatically executed or called when objects of that class will be created. As when object or derived class is defined automatically derived class constructor and base class constructor will execute inside them.

Statement (d) : Overloading is a type of polymorphism. This statement is correct. Operator is a type of polymorphism in which an operator is overloaded to give a different meaning or user defined meaning. It allows to change the way an operator can work.

72. Which of the following are legal statements in C programming language?

- (A) `int * P = &44;`
- (B) `int * P = &r;`
- (C) `int P = &a;`
- (D) `int P = a;`

Choose the correct option:

- (a) (A) and (B)
- (b) (B) and (C)
- (c) (B) and (D)
- (d) (A) and (D)

Ans. (c) :

a - Pointer variable P cannot store address of a value which is not assigned to a variable.

b - Pointer variable P can store address of variable r.

c - Integer variable P cannot store address of variable a.

d - Integer variable P can store value of variable a.

Hence `int*P = & r ;` and `int P = a;` are legal statements in C programming language.

73. Two concurrent executing transactions T_1 and T_2 are allowed to update same stock item say 'A' in an uncontrolled manner. In such scenario, following problems may occur:

- (A) Dirty read problem
- (B) Lose update problem
- (C) Transaction failure
- (D) Inconsistent database state

Which of the following option is correct if database system has no concurrency module and allows concurrent execution of above two transactions?

- (a) (A), (B) and (C) only
- (b) (C) and (D) only
- (c) (A) and (B) only
- (d) (A), (B) and (D) only

Ans. (d) : Problem & that can occur during concurrent execution of two transactions in an uncontrolled manner.

(a) Lost update problem: Lost update problem occurs when two transactions that access the same database items have their operations interleaved in a way that makes the value of some database item incorrect.

(b) Dirty read Problem: It occurs when a transaction is allowed to retrieve or update a record that has been updated by another transaction. Because it has not yet been committed there is always a possibility that it will never be committed but rather roll back.

(c) **Inconsistent database state:** It occurs when a transaction reads several values but a second transaction updates some of these values during the execution of first. This causes an inconsistency.

For example: If one transaction is calculating an aggregate summary function on a number of records while other transactions are updating some of records.

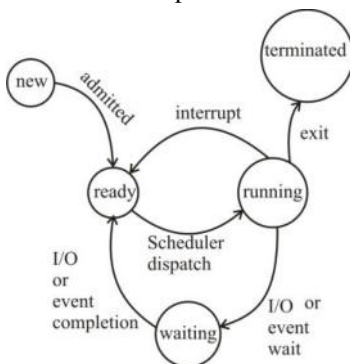
74. Identify the circumstances under which pre-emptive CPU scheduling is used:

- (A) A process switches from Running state to Ready state
- (B) A process switches from Waiting state to Ready state
- (C) A process completes its execution
- (D) A process switches from Ready to Waiting state

Choose the correct option:

- (a) (A) and (B)
- (b) (A) and (D)
- (c) (C) and (D)
- (d) (A), (B), (C) only

Ans. (a) : To understand pre-emptive scheduling we first need to understand the process state diagram.



Scheduling is performed from when the process is taken from ready state on to the running state.

Hence CPU scheduling decisions take place under one of four conditions.

- (1) Process switch from running state to waiting state.
- (2) Process switch from running state to Ready State.
- (3) Process switch from waiting state to ready state (statement b)
- (4) Process termination.

Statement C- No need for preemptive scheduling as execution is complete.

Statement D- Process cannot switch from Ready state to waiting state.

75. The following multithreaded algorithm computes transpose of a matrix in parallel:

p Trans (X, Y, N)

if N = 1

then Y[1, 1] ← X [1, 1]

else partition X into four $(N/2) \times (N/2)$ sub matrices $X_{11}, Y_{12}, Y_{21}, Y_{22}$

spawn p Trans $(X_{11}, Y_{11}, N/2)$

spawn p Trans $(X_{12}, Y_{12}, N/2)$

spawn p Trans $(X_{21}, Y_{21}, N/2)$

spawn p Trans $(X_{22}, Y_{22}, N/2)$

What is the asymptotic parallelism of the algorithm?

- (a) T_1/T_∞ or $\theta(N^2/\lg N)$
- (b) T_1/T_∞ or $\theta(N/\lg N)$
- (c) T_2/T_∞ or $\theta(\lg N/N^2)$
- (d) T_1/T_∞ or $\theta(\lg N/N)$

Ans. (a) : During transpose of the matrix, it is partitioned into 4 submatrix of size $N/2$.

If $n = 1$, then $T(1) = 1$

Else $T_1(n) = 4T_1(N/2) + O(1)$ (i)

solving equation 1 by master's theorem:

$\backslash(\{n^{\log b^a}\} = \backslash;\{n^{\log 2^4}\} = \{n^2\})$,

here $a = 4$ and $b = 2$

so $T_1(n) = O(N^2)$

But algorithm is multithreaded and transpose is going in parallel and we solved one sub problem already.

So, $T_n(N) = T_n(N/2) + O(1)$

$T_n(N)$ becomes $O(\log N)$

asymptotic parallelism of the algorithm

T_1/T_∞ or $\theta(N^2/\log N)$

76. Consider the following statements:

- (A) The running time of dynamic programming algorithm is always $\theta(\rho)$ where ρ is number of sub problems.
- (B) When a recurrence relation has cyclic dependency. It is impossible to use that recurrence relation (unmodified) in a correct dynamic program.
- (C) For a dynamic programming algorithm computing all values in a bottom-up fashion is asymptotically faster than using recursion and memorization.
- (D) If a problem X can be reduced to a known NP hard problem then X must be NP hard.

Which of the statement(s) is (are) true?

- (a) Only (B) and (A)
- (b) Only (B)
- (c) Only (B) and (C)
- (d) Only (B) and (D)

Ans. (b) : (a) The running time of dynamic programming algorithm is always θ

(ρ) where P is number of sub problems.

This statement is incorrect running time for dynamic algorithms is the number of sub-problems multiplies with time for each sub-problem. It means running time of dynamic programming algorithms is $O(1)$.

(b) When a recurrence relation has cyclic dependency it is impossible to use that recurrence relation (unmodified) in a correct dynamic program.

Cyclic dependency is relation between two or more modules which depends on each other indirectly or directly, these are mutually recursive. So, given statement is correct.

(c) For a dynamic programming algorithms, computing all values in a bottom up fashion is asymptotically faster than using recursion and memorization.

This statement is incorrect for a dynamic programming algorithm, computing all values using recursion and memorization takes less time than computing in bottom up fashion.

(d) If a problem X can be reduced to known NP-hard problem, then X must be NP hard.

If a problem X can be reduced to a known NP hard problem, then X must be NP complete not NP hard.

So given statement is incorrect.

77. Consider the following statements:

(A) Fiber optic cable is much lighter than copper cable

(B) Fiber optic cable is not affected by power surges or electromagnetic interference.

(C) Optical transmission is inherently bidirectional

Which of the statements is (are) correct?

(a) Only (A) and (B)

(b) Only (A) and (C)

(c) Only (B) and (C)

(d) (A), (B) and (C)

Ans. (a) : only (A) and (B) is correct→

(a) Fiber optic cable is much lighter than copper cable. Given statement is correct. Fiber optic cable transmit signal in the form of light pulses. they can transmit data at higher speed than copper cables and these are lighter than copper cables. Also fiber optic cable have higher bandwidth.

(b) Fiber optic cable is not affected by power surges or electromagnetic interference.

This statement is correct fiber optic cables are non metallic and they transmit signal in the form of light pulses. that's why they are immune to electromagnetic interference. Integrity of signal is not affected by electrical noise.

(c) Optical transmission is inherently bidirectional. This statement is incorrect. Two optical modules are connected by single fiber cable which supports unidirectional communication for bidirectional communication, two cables will be required.

78. Consider the following statements:

(A) Windows Azure is a cloud based operating system

(B) Google App Engine is an integrated set of online services for consumers to communicate and share with others.

(C) Amazon Cloud Front is a web service for content delivery.

Which of the statements is (are) correct?

- (a) Only (A) and (B)
- (b) Only (A) and (C)
- (c) Only (B) and (C)
- (d) (A), (B) and (C)

Ans. (b) : (a) Window Azure is a cloud-based operating system.

This statement is correct. Windows Azure is a cloud based operating system which enables running of business applications, services and work load in the cloud itself. It mainly deals with the foundation of running applications and keeping the data on the cloud.

(b) Google App Engine- is an integrated set of online services for consumers to communicate and share with others. Google app engine is PaaS(platform as a service) product. It provides web developing services to developers. Apps must be written in Java or python. So this statement is incorrect.

(c) Amazon Cloud Front- is a web service content delivery. Amazon cloud front is a fast content delivery network service that securely delivers data, video applications and APIs to customers. Cloud front is integrated with AWS global infrastructure and services. So statement is correct.

79. Consider the following statements with respect to network security:

- (A) Message confidentiality means that the sender and the receiver expect privacy.**
- (B) Message integrity means that the data must arrive at the receiver exactly as they were sent.**
- (C) Message authentication means the receiver is ensured that the message is coming from the intended sender.**

Which of the statements is (are) correct?

- (a) Only (A) and (B)
- (b) Only (A) and (C)
- (c) Only (B) and (C)
- (d) (A), (B) and (C)

Ans. (d) : (a) message confidentiality means that the sender and the receiver expect privacy.

Message confidentiality means that only sender and intended receiver should be able to access the contents of a message.

Confidentiality gets compromised if an unauthorized person is able to access a message. It means sender and receiver expect privacy.

(b) Message integrity means that the data must arrive at the receiver exactly as they were sent.

When the contents of a message are changed after the sender sends it, but before it reaches the intended

recipient, we say that the integrity of the message is lost. It means that the data must arrive at the receiver exactly as they were sent.

(c) Message authentication means the receiver is ensured that the message is coming from the intended sender.

Authentication mechanisms help establish proof of identities. The authentication process ensures that the origin of an electronic message or document is correctly identified. It means that receiver is ensured that the message is coming from the intended sender.

80. Consider the following:

(A) Trapping at local maxima

(B) Reaching a plateau

(C) Traversal along the ridge

Which of the following option represents shortcomings of the hill climbing algorithm?

(a) (A) and (B) only

(b) (A) and (C) only

(c) (B) and (C) only

(d) (A), (B) and (C)

Ans. (d) : HILL CLIMBING ALGORITHM:- In hill climbing algorithm at each point in the search path a successor node that appears to lead most quickly to the top of the hill or goal is selected for exploration.

In this method of searching, the generate and test method is augmented by a heuristic foundation which measures the closeness of the current state to the goal state.

Explanation:- Problems in hill claiming are:

(1) Local maximum: It is a state which is better than of its neighbors but is not better than some other states which are farther away. Solution of this problem is backtracking.

(2) Plateau: It is a that area of search space in which a whole set of neighboring states have the same value. On a plateau, it is not possible to determine the best direction in which to move.

(3) Ridge:- It is an area of search space which is higher them the surrounding areas and that itself has a slope.

81. Consider the following learning algorithms:

(A) Logistic repression

(B) Back propagation

(C) Linear repression

Which of the following option represents classification algorithms?

(a) (A) and (B) only

(b) (A) and (C) only

(c) (B) and (C) only

(d) (A), (B) and (C)

Ans.(*): LEARNING ALGORITHM:- These algorithms are used in machine learning to help technology in human learning process. It is used to process data to extract patterns appropriate for application in a new system.

From above given options, only (A) and (B) are learning algorithms.

(a) Logistic regression: It is used for binary classification problems. It is used to examine and describe the relationship between a binary variable and set of predictor variables. The primary objective of logistic regression is to model the mean of the response variables, given a set of predictor variables.

(b) Back Propagation:- It is the essence of neural net training. It is the method of fine tuning the weights of a neural net based on the error rate obtained in the previous iteration. It is a standard method of training artificial neural network.

82. Match List-I and List-II

| List-I | | List-II | |
|--------|--------------------|---------|--|
| A. | Physical layer | 1. | Provide token management service |
| B. | Transport layer | 2. | Concerned with transmitting raw bits over a communication channel |
| C. | Session layer | 3. | Concerned with the syntax and semantics of the information transmitted |
| D. | Presentation layer | 4. | True end-to-end layer from source to destination |

- Choose the correct option from those given below:**
- (a) A-3, B-4, C-3, D-1
 - (b) A-4, B-3, C-2, D-1
 - (c) A-2, B-4, C-1, D-3
 - (d) A-4, B-2, C-1, D-3

| | | | |
|-------------------|-----------------|----------------|---|
| Ans. (c) : | | | |
| List-I | | List-II | |
| A. | Physical layer | 2. | Concerned with transmitting raw bits over a communication channel |
| B. | Transport layer | 4. | True end-to-end layer from source to destination |

| | | | |
|----|--------------------|----|--|
| C. | Session layer | 1. | Provide token management service |
| D. | Presentation layer | 3. | Concerned with the syntax and semantics of the information transmitted |

83. According to the ISO-9126 Standard Quality Model. Match the attributes given in List-I with their definition in List-II:

| List-I | | List-II | |
|--------|-----------------|---------|---|
| A. | Functionality | 1. | Relationship between level of performance and amount of resources |
| B. | Reliability | 2. | Characteristics related with achievement of purpose |
| C. | Efficiency | 3. | Effort needed to make for improvement |
| D. | Maintainability | 4. | Capability of software to maintain performance of software |

Choose the correct option from the ones given below:

- (a) A-1, B-2, C-3, D-4
- (b) A-2, B-1, C-4, D-3
- (c) A-2, B-4, C-1, D-3
- (d) A-1, B-2, C-4, D-3

| Ans. (c) : | | | |
|-------------------|-----------------|---------|---|
| List-I | | List-II | |
| A. | Functionality | 2. | Characteristics related with achievement of purpose |
| B. | Reliability | 4. | Capability of software to maintain performance of software |
| C. | Efficiency | 1. | Relationship between level of performance and amount of resources |
| D. | Maintainability | 3. | Effort needed to make for improvement |

84. Match the Agile Process models with the task performed during the model:

| List-I | | List-II | |
|--------|-------------------------------|---------|---|
| A. | Scrum | 1. | CRC cards |
| B. | Adaptive software development | 2. | Sprint backlog |
| C. | Extreme programming | 3. | <action> the
<result>
<by/for/of/to>
a(n) <object> |
| D. | Feature-driven development | 4. | Time box
release plan |

Choose the correct option from those given below:

- (a) A-2, B-4, C-1, D-3
- (b) A-1, B-3, C-2, D-4
- (c) A-2, B-1, C-4, D-3
- (d) A-1, B-4, C-2, D-3

Ans. (a) :

| List-I | | List-II | |
|--------|-------------------------------|---------|---|
| A. | Scrum | 2. | Sprint backlog |
| B. | Adaptive software development | 4. | Time box release plan |
| C. | Extreme programming | 1. | CRC cards |
| D. | Feature-driven development | 3. | <action> the
<result>
<by/for/of/to> a(n)
<object> |

85. Match List-I with List-II

| List-I | | List-II | |
|--------|------------------|---------|--|
| A. | Frame attribute | 1. | to create link in HTML |
| B. | <tr>tag | 2. | for vertical alignment of content in cell |
| C. | Valign attribute | 3. | to enclose each row in table |
| D. | <a>tag | 4. | to specify the side of the table frame that display border |

Choose the correct option from those given below:

- (a) A-1, B-2, C-3, D-4
- (b) A-2, B-1, C-3, D-4
- (c) A-4, B-3, C-2, D-1
- (d) A-3, B-4, C-2, D-1

Ans. (c) :

| List-I | | List-II | |
|---------------|------------------|----------------|--|
| A. | Frame attribute | 4. | to specify the side of the table frame that display border |
| B. | <tr>tag | 3. | to enclose each row in table |
| C. | Valign attribute | 2. | for vertical alignment of content in cell |
| D. | <a>tag | 1. | to create link in HTML |

- 86. An Instruction is stored at location 500 with its address field at location 501. The address field has the value 400. A processor register R_1 contains the number 200. Match the addressing mode (List-I), given below with effective address (List-II) for the given instruction:**

| List-I | | List-II | |
|---------------|--|----------------|------------|
| A. | Direct | 1. | 200 |
| B. | Register indirect | 2. | 902 |
| C. | Index with R_1 as the index register | 3. | 400 |
| D. | Relative | 4. | 600 |

Choose the correct option from those given below:

- (a) A-3, B-1, C-4, D-2
 (b) A-1, B-2, C-3, D-4
 (c) A-4, B-2, C-3, D-1
 (d) A-4, B-3, C-2, D-1

Ans. (a) :

| List-I | | List-II | |
|---------------|--|----------------|-----|
| A. | Direct | 3. | 400 |
| B. | Register indirect | 1. | 200 |
| C. | Index with R_1 as the index register | 4. | 600 |
| D. | Relative | 2. | 902 |

- 87. Match List-I and List-II**

| List-I | | List-II | |
|---------------|--------------------------|----------------|--|
| A. | Isolated I/O | 1. | same set of control signal for I/O and memory communication |
| B. | Memory mapped I/O | 2. | separate instructions for I/O and memory communication |
| C. | I/O interface | 3. | requires control signals to be transmitted between the |

| | | | |
|-----------|-----------------------------------|-----------|--|
| | | | communication units |
| D. | Asynchronous data transfer | 4. | resolve the differences in central computer and peripherals |

Choose the correct option from those given below:

- (a) A-2, B-3, C-4, D-1
- (b) A-1, B-2, C-3, D-4
- (c) A-2, B-1, C-4, D-3
- (d) A-1, B-2, C-4, D-3

Ans. (c) :

List-I

List-II

| | | | |
|----|----------------------------|----|--|
| A. | Isolated I/O | 2. | separate instructions for I/O and memory communication |
| B. | Memory mapped I/O | 1. | same set of control signal for I/O and memory communication |
| C. | I/O interface | 4. | resolve the differences in central computer and peripherals |
| D. | Asynchronous data transfer | 3. | requires control signals to be transmitted between the communication units |

88. Match List-I with List-II:

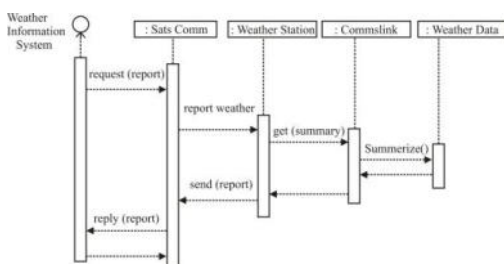
| List-I | | List-II | |
|---------------|--------------------------------------|----------------|---|
| A. | Micro operation | 1. | Specify micro operations |
| B. | Micro programmed control unit | 2. | Improve CPU utilization |
| C. | Interrupts | 3. | Control Memory |
| D. | Micro instruction | 4. | Elementary operation performed on data stored in registers |

Choose the correct option from those given below:

- (a) A-4, B-3, C-2, D-1
- (b) A-4, B-3, C-1, D-2
- (c) A-3, B-4, C-1, D-2
- (d) A-3, B-4, C-2, D-1

| Ans. (a) : | | | |
|------------|-------------------------------|---------|--|
| List-I | | List-II | |
| A. | Micro operation | 4. | Elementary operation performed on data stored in registers |
| B. | Micro programmed control unit | 3. | Control Memory |
| C. | Interrupts | 2. | Improve CPU utilization |
| D. | Micro instruction | 1. | Specify micro operations |

89.



The sequence diagram given in Figure for the Weather Information System takes place. When an external system requests the summarized data from the weather station. The me easing order of lifeline for the objects in the system are:

- Sats Comm → Weather station → Commlink → Weather data
- Sats comms → Comms link → Weather station → Weather data
- Weather data → Comms link → Weather station → Sat Comms
- Weather data → Weather station → Comms link → Sat Comms

Ans. (c) : In this scenario, first connection request is established, weather information system is requesting for the weather report.

Request takes place as:

- Request will be sent from weather information system to sat communication.
- Sat communication is forwarding that request to weather station.
- Then weather station sends get (summary) request to communication link.
- After that, it will fetch data from weather data.

Reply Process:

- As the communication link is nearest to weather data, first it will get the data.
- Then it goes from communication link to weather station.
- Then weather station to sat comm.

(4) At the end, weather information system get the weather information.

So increasing order of lifeline of objects in the system will be same as objects similar or larger distance from weather data, in the way reply is gone through. So it will be.

Weather data → Comms link → Weather station → Sat Comms.

90. The Data Encryption Standard (DES) has a function consists of four steps. Which of the following is correct order of these four steps?

- (a) an expansion permutation. S-boxes an XOR operation a straight permutation
- (b) an expansion permutation an XOR operation S-boxes a straight permutation
- (c) a straight permutation S-boxes an XOR operation an expansion permutation
- (d) a straight permutation an XOR operation, S-boxes an expansion permutation

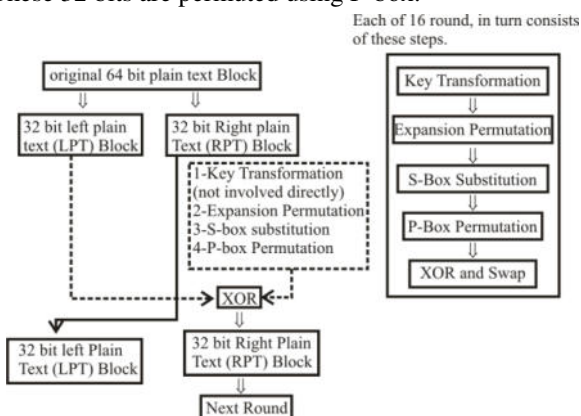
Ans. (b) : Data Encryption standard (DES)- Data encryption standard (DES) is a block cipher that encrypts data in blocks of size 64 bits each. That is 64 bits of plain text goes input to the DES which produces 64 bit of cipher text key length is 56bits.

Explanation:

Broad level steps of DES are:

- In first step 64 bit plain text block is handed over to initial permutation.
- Next initial permutation (IP) produces two halves of permuted block, say left plain text and right plain text.
- Now each LPT and RPT go through 16 rounds of encryption process each with its own key.
- At the end, LPT and RPT are rejoined and a final permutation is performed on combined block.

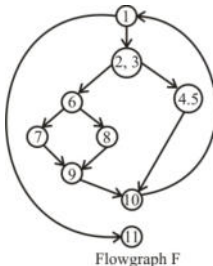
During expansion permutation, it goes to S-box substitution after doing XOR of 48 bit key with 48 bit right plain text. S-Box produces the 32 bit output using substitution technique. A a straight permutation is done. These 32 bits are permuted using P-box.



Question Numbers : (91 to 95)

Question Label: Comprehension

A flow graph F with entry node (1) and exit node (11) is shown below:



Flowgraph F

91. How many nodes are there in flowgraph F?

- (a) 9 (b) 10
(c) 11 (d) 12

Ans. (a): In given graph contains 9 nodes and 11 edges. The nodes are nothing but vertices. Here, In given diagram the number of rounds one the nodes. The nodes one 1, (2, 3), (4, 5), 6, 7, 8, 9, 10, 11

Count of total nodes in flow graph F are 9.

92. What is the cyclomatic complexity of flowgraph F?

- (a) 2 (b) 3
(c) 4 (d) 5

Ans. (c) : Cyclomatic complexity of a flow graph is the quantitative measure of the number of linearly independent paths in it.

for calculation of cyclomatic complexity, it is necessary to understand some parameters.

1-Region- In out case, there are 4 regions.

2-Edges- In out case, there are 9 nodes.

3-Nodes- In our case, there is 1 connected component.

5-Predicates- These are nodes that contain condition in our case, there are 3 predicates namely node 1, 2, 3, 6.

Formula- There are three ways of calculating the cyclamate complexity m,

1- $m = \text{number of regions}$

2- $m = E - N + 2P$

where

E - Number of Edges

N - Number of Nodes

P - Number of connected component

$m = P + 1$

3- where, P = number of predicate nodes

Calculation-

using formula 1, $m = 4$

using formula 2, $m = 11 - 9 + 2(1) = 4$

using formula 3, $m = 3 + 1 = 4$

93. How many regions are there in flow graph F?

- (a) 2 (b) 3
(c) 4 (d) 5

Ans. (c) : REGION- region is an area that is bounded by edges and cannot be further subdivided. Region can be finite nature and can also be infinite nature planar graph can only have one infinite region.

The graph in the data has 3 finite or closed regions.

Region 1- Bounded by edge 1-10-1

Region 2- Bounded by edges (2.3)-6-8-9-10-(4.5)-(2.3)

Region 3- Bounded by edges 6-7-9-8-6

The fourth region is infinite or open region Region 4-1-11

94. How many nodes are there in the longest independent path?

- (a) 6 (b) 7
(c) 8 (d) 9

Ans. (c) : Path 1: 1-11

Path 2: 1-2-3-4-5-10-1-11

Path 3: 1-(2, 3)-6-8-9-10-1-11

Path 4: 1-(2, 3)-6-7-9-10-1-11

Each of the above path introduces a new edge and Path 3 and Path 4 one longest independent paths with 8 Node.

Note that a path like 1-(2, 3)-4-5-10-1-2-3-6-8-9-10-1-11 is not an independent path because it includes already specified edges and doesn't exclusively add new edges.

95. How many predicate nodes are there and what are their names?

- (a) Three : (1, (2, 3) , 6)
(b) Three : (1, 4, 6)
(c) Three : ((2, 3) , 6, 10, 11)
(d) Four: ((2, 3) , 6, 9, 10)

Ans. (a) : Predicate node is a node that contains a condition. This node has 2 or more outgoing edges from it.

In the given graph the nodes with a condition or with 2 or more outgoing edges one - 1, 2.3, 6.

Hence, there are 3 predicates nodes.

Question numbers: (146 to 150)

Question Label: Comprehension

An organization needs to maintain database having five attributes A, B, C, D, E, These attributes are functionally dependent on each other for which functional dependency set F is given as : $F : \{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$. Consider a universal relation R(A, B, C, D, E) with functional dependency set F. Also all attributes are simple and take atomic values only.

96. Minimal cover F' of functional dependency set F is:

- (a) $F' = \{A \rightarrow B, A \rightarrow C, BC \rightarrow D, D \rightarrow E\}$
(b) $F' = \{A \rightarrow BC, B \rightarrow D, D \rightarrow E\}$
(c) $F' = \{A \rightarrow B, A \rightarrow C, A \rightarrow D, D \rightarrow E\}$
(d) $F' = \{A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D, D \rightarrow E\}$

Ans. (a) : A canonical cover is simplified and reduced version of the given set of functional dependencies minimal canonical cover is a cover for which removed of any single functional dependency destroys the covering property.

Data-

The functional dependencies over the universal relation R(A, B, C, D, E) are

$A \rightarrow BC = A \rightarrow B, A \rightarrow C$

$D \rightarrow E$

$BC \rightarrow D$

$A \rightarrow D$

Method- Following are the steps to find the minimal cover.

1- Write all FDs such that RHS of each FD has only one attribute.

$A \rightarrow B$

$A \rightarrow C$

$D \rightarrow E$

$BC \rightarrow D$

$A \rightarrow D$

2- Remove transitive FDs and any other FDs where the RHS can be driven of LHS of the other FDs. In this case, $A \rightarrow D$ is a redundant FD.

Hence the result becomes

$A \rightarrow B$

$A \rightarrow C$

$D \rightarrow E$

$BC \rightarrow D$

3- Finding the FDs with 2 attributes in LHS, we get the dependency $BC \rightarrow D$. Try to eliminate the attributes using closure of both the attribute B and C.

In this case, it is not possible to remove either B or C because both have different closures.

Hence, minimal cover is

$A \rightarrow B$

$A \rightarrow C$

$D \rightarrow E$

$BC \rightarrow D$

97. Identify primary key of table R with functional dependency set F

(a) BC

(b) AD

(c) A

(d) AB

Ans. (c) : A primary Key is a special relational database table column (or combination of columns) designated to uniquely identify all table records.

Data- In the functional dependency set f of relation R, the key 'A' is not in the RHS of any FD.

Explanation:- It A is not in RHS of any FD, it is mandatory to includes it in the set of candidate key.

Now, to make sure if 'A' alone is the primary key, we write its closure. The set of all those attributes which can be functionally determined from an attribute set is called as a closure of that attribute set.

$A^+ = \{A, B, C, D, E\}$

A derives all the attributes of relation R. Hence, it is the primary key of R.

98. Identify the normal form in which relation R belong to

(a) 1 NF

(b) 2 NF

(c) 3 NF

(d) BCNF

Ans. (b) : Normal forms refer to type of functional dependencies.

1NF- A relation is in 1 NF if it contains an atomic value.

2NF- A relation will be in 2NF if it is in 1NF and all non-key attributes are fully functional dependent on the primary key.

3NF- A relation will be in 3NF if it is in 2NF and no transitive dependency exists.

BCNF- A relation is in BCNF if every functional dependency $X \rightarrow Y$, X is the super key of the table.

Data

1- A is the primary key.

2- There is no partial dependency in R. Partial dependency occurs when a non-prime attribute is functionally dependent on part of a candidate key.

Hence, R, is in 2NF

Explanation- R is in 2NF and not in 3NF because $D \rightarrow E$, $BC \rightarrow D$ neither have a super key in their LHS nor do they have a primary key in their RHS.

If it is not in 3NF, it cannot be in BCNF.

99. Identify the redundant functional dependency in F

(a) $BC \rightarrow D$

(b) $D \rightarrow E$

(c) $A \rightarrow D$

(d) $A \rightarrow BC$

Ans. (c) : Dependency $A \rightarrow D$ is derived from $A \rightarrow BC$, $BC \rightarrow D$. Hence it is transitive and redundant.

100. Assume that given table R is decomposed in two tables

$R_1(A, B, C)$ with functional dependency set $f_1 = \{A \rightarrow B, A \rightarrow C\}$ and $R_2(A, D, E)$ with FD set $F_2 = \{A \rightarrow D, D \rightarrow E\}$.

Which of the following option is true w.r.t given decomposition?

(a) Dependency preservation property is followed

(b) R_1 and R_2 are both in 2 NF

(c) R_2 is in 2 NF and R_3 is in 3 NF

(d) R_1 is in 3 NF and R_2 is in 2 NF

Ans. (d) : Relation $R(A, B, C, D, E)$ with its functional dependencies $F: (A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D)$ has been decomposed in two relations.

1- $R_1(A, B, C)$ with functional dependency set $f_1 = \{A \rightarrow B, A \rightarrow C\}$

2- $R_2(A, D, E)$ with FD set $F_2 = \{A \rightarrow D, D \rightarrow E\}$

Dependency Preservation- In both the relations the dependency

$BC \rightarrow D$ cannot be determined.

Hence both R_1 and R_2 are not following dependency preservation property.

Normal Forms:- In R_1 , A is the candidate key. and the functional dependency set does not contain any transitive partial dependency. Hence the set is in 3NF.

In R_1 , A is the candidate key but the FD $D \rightarrow E$ breaks the rules of 3NF. Hence, it is in 2NF.